
Safe Until It Isn't: Correlation Spikes, Portfolio Fragility, and True Diversification

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Abstract

This paper examines how dependence across financial assets changes between calm and stressed markets. We study daily returns for 14 exchange-traded funds covering equities, bonds, commodities, gold, oil, and real estate. A two-state Markov-switching model on monthly S&P 500 returns identifies the primary regimes, validated against a VIX-and-drawdown rule and a weekly specification. Regime-conditional correlations are evaluated using bootstrap inference, principal component analysis, and a Forbes–Rigobon volatility adjustment.

The results are panel-dependent. Average correlation rises strongly among international equities but falls slightly across asset classes, where tighter risk-asset comovement is offset by lower correlations involving government bonds and gold. Nevertheless, both panels become more concentrated in stress, with a stronger dominant component and fewer effective independent directions. After adjustment, no pair retains a robust correlation breakdown, indicating that the raw increases largely reflect higher stress-period volatility. Diversification therefore weakens most clearly within international equities, while sovereign bonds and gold remain comparatively resilient.

Keywords: asset correlation; market stress; Markov switching; bootstrap inference; principal component analysis; Forbes–Rigobon adjustment; diversification

Abbreviations: ETF, exchange-traded fund; PCA, principal component analysis; VIX, volatility index

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1 Introduction

Diversification is one of the central principles of financial decision-making. An investor does not evaluate an asset only by its expected return or volatility, but also by how it moves relative to the other assets held alongside it. The insight formalized by Markowitz (1952) is simple but consequential: portfolio risk depends on the joint behavior of returns. Holding more assets is therefore not sufficient by itself; the benefit comes from combining exposures whose returns do not move together.

Consider a portfolio of N risky assets with weights $w_1, \dots, w_N$, where $\sum_{i=1}^N w_i = 1$. Its return is the weighted sum of the individual asset returns:

$$R_{p,t} = \sum_{i=1}^N w_i R_{i,t} = \mathbf{w}^\top \mathbf{R}_t, \quad (1)$$

and its expected return is

$$\mathbb{E}[R_{p,t}] = \sum_{i=1}^N w_i \mathbb{E}[R_{i,t}] = \mathbf{w}^\top \boldsymbol{\mu}. \quad (2)$$

Expected portfolio return is therefore linear in the weights: each asset contributes in direct proportion to the amount invested in it.

Portfolio risk is different. When risk is measured by return variance,

$$\begin{aligned} \sigma_p^2 &= \text{Var}(R_{p,t}) = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \\ &= \sum_{i=1}^N w_i^2 \sigma_i^2 + 2 \sum_{i < j} w_i w_j \rho_{ij} \sigma_i \sigma_j, \end{aligned} \quad (3)$$

where σ_i^2 is the variance of asset i and ρ_{ij} is the correlation between assets i and j . Risk is not a weighted average of individual volatilities. It also contains every pairwise interaction in the portfolio. As the number of assets grows, these covariance terms become increasingly important relative to any single variance. Appendix A makes this result explicit for an equally weighted $1/N$ portfolio, a simple allocation rule whose empirical performance relative to optimized strategies is studied by DeMiguel et al. (2009). Two portfolios containing assets with similar standalone volatility can consequently have very different risk if their correlation structures differ.

This distinction explains both the value and the fragility of diversification. Low or negative correlations allow losses in one exposure to be offset by different behavior elsewhere. Yet correlations are not fixed parameters. During market stress, common shocks, deleveraging, liquidity needs, and shifts in risk appetite can cause assets that appeared distinct in calm conditions to move together. A correlation matrix estimated over the full sample

may then describe neither calm nor crisis particularly well: it averages across environments in which the economic relationships between assets can differ. The problem is especially important because diversification is most valuable precisely when aggregate losses are large.

The project therefore focuses on the stability of dependence rather than on forecasting returns or constructing an optimal allocation. It studies 14 exchange-traded funds spanning equities, government and corporate bonds, commodities, gold, oil, and real estate. The assets form two complementary perspectives: a cross-asset panel asks whether different sources of market exposure remain distinct under stress, while an international-equity panel asks whether geographic diversification survives when global equity markets fall together.

The central research question is: *How does the dependence structure across major asset classes and international equity markets change between calm and stress regimes?* Answering it requires more than observing that a sample correlation is higher in one period. Estimated changes may reflect sampling uncertainty, may be concentrated in only a few pairs, or may arise mechanically because volatility is higher in stress. Moreover, a collection of pairwise estimates can miss a broader collapse toward one dominant market factor. The purpose of the analysis is therefore diagnostic: to determine where diversification weakens, whether the change is statistically credible, whether market-wide dependence becomes more concentrated, and how much of the apparent breakdown remains after accounting for stress-period volatility.

By separating calm and stress conditions, the study provides a clearer account of when apparently different assets continue to offer distinct risk exposures and when they cease to do so. This matters even without implementing a portfolio strategy, because any use of historical covariance estimates rests on an assumption about how stable those relationships will remain when market conditions change.

2 Literature and previous research

The introduction has shown that portfolio diversification depends on the stability of the relationships between assets. However, empirical research indicates that correlations are not constant over time and may change substantially during periods of financial stress. An asset that appears to provide diversification under normal conditions may therefore behave differently when markets experience high volatility and falling prices.

Longin and Solnik (2001) show that correlations between international equity markets increase particularly during severe downturns. Ang and Chen (2002) document a simi-

lar asymmetry for U.S. equity portfolios, where correlations are stronger after negative market movements than after positive movements of comparable size. These findings imply that unconditional correlations can understate the comovement investors face in adverse states. For a diversification analysis, it is therefore not enough to ask whether assets appear diversified on average. The relevant question is whether those relationships remain stable when markets are under stress.

A defensible answer first requires a systematic definition of market stress. Historical crisis dates can be selected manually, but such choices are partly subjective and may bias the results. Markov-switching models offer a statistical alternative by allowing returns to move between latent states with different means and variances (Hamilton, 1989). This approach is well suited to financial markets because stress is usually characterized by high volatility, lower returns, and persistent regime dynamics. Prior work shows that such regimes matter for asset allocation because return distributions and dependence structures can differ substantially across states (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007).

However, higher stress-period correlations must be interpreted carefully. Forbes and Rigobon (2002) show that correlations can increase mechanically when volatility rises, even if the underlying relationship between markets has not changed. They distinguish between contagion, meaning a genuine strengthening of linkages after a shock, and interdependence, meaning strong existing relationships observed in a more volatile environment. This distinction is central for our paper. Raw correlations describe the comovement that investors actually experience, while Forbes–Rigobon-adjusted correlations ask whether the increase remains after correcting for heteroskedasticity. A stress correlation increase can therefore signal a realized loss of diversification without necessarily proving structural contagion.

Pairwise correlations also provide only a partial view of diversification. Even if some pairs remain stable, the portfolio as a whole may become more exposed to a common risk factor. Principal component analysis captures this broader concentration by measuring how much of total variation is explained by the leading components (Jolliffe, 2002). A higher first principal-component share in stress means that returns are increasingly driven by one common direction, leaving fewer independent sources of risk. This idea is closely related to the systemic-risk literature: Kritzman et al. (2011) interpret a rising concentration of variance in a few components as an absorption ratio, and Billio et al. (2012) show that financial-system connectedness tightens during crises.

This study combines these strands of research in one empirical framework. It compares regime-conditional correlations for a cross-asset universe and an international equity universe, tests their sampling uncertainty, measures portfolio-level concentration with PCA,

and applies the Forbes–Rigobon adjustment to distinguish volatility-driven comovement from stronger evidence of contagion.

3 Data

3.1 Asset Universe and the Case for ETFs

This study uses exchange-traded funds (ETFs) as investable proxies for broad market segments rather than constructing synthetic portfolios from individual securities. This choice is deliberate for three reasons. First, ETFs map into economically interpretable portfolio sleeves such as U.S. equities, Treasuries, investment-grade bonds, high-yield credit, gold, broad commodities, oil, and listed real estate because the selected products have explicitly stated benchmark indexes or investment objectives documented by their issuers (State Street Global Advisors, 2026b; iShares, 2026h,i,a,g; State Street Global Advisors, 2026a; Invesco, 2026; USCF Investments, 2026; Vanguard, 2026). Second, ETFs are directly tradable market vehicles: ETF shares trade intraday on exchanges at market-determined prices, while the creation-redemption mechanism and portfolio transparency help keep ETF prices linked to the value of the underlying holdings (Investment Company Institute, 2026; Madhavan, 2016). This makes the empirical design closer to the portfolio problem faced by investors who allocate across tradable market exposures. Third, a fixed ETF universe avoids the additional firm-level reconstruction and rebalancing choices that arise when researchers hand-assemble changing baskets of individual securities *ex post*. At the same time, this design does *not* fully eliminate product-level survivorship concerns, because the selected ETFs are surviving vehicles observed through the sample end. Survivorship bias is a known issue in empirical performance studies because excluding disappeared funds or securities can bias measured performance and risk characteristics (Brown et al., 1992; Elton et al., 1996).

The analysis is conducted on two panels. The first panel is the main *cross-asset* universe and is designed to span the canonical building blocks of a diversified portfolio. The second panel is an *international equity* universe and is designed to examine whether geographic equity diversification remains effective once market stress emerges.

Panel A: Cross-asset universe. The cross-asset panel contains nine ETFs:

$$\{\text{SPY}, \text{SHY}, \text{TLT}, \text{AGG}, \text{HYG}, \text{GLD}, \text{DBC}, \text{USO}, \text{VNQ}\}.$$

The economic role of each ETF is summarized in Appendix Table 6.

The cross-asset panel is intentionally *parsimonious rather than exhaustive*. The purpose is not to span every investable niche, but to cover the main dimensions along which diversification is usually justified in practice: equity risk, sovereign duration, aggregate fixed income, risky credit, precious metals, broad commodities, energy, and real estate. SPY provides broad U.S. equity exposure through the S&P 500 Index (State Street Global Advisors, 2026b); SHY and TLT span the short and long ends of the U.S. Treasury curve (iShares, 2026h,i); AGG captures the U.S. investment-grade aggregate bond market (iShares, 2026a); HYG provides exposure to U.S. dollar-denominated high-yield corporate bonds (iShares, 2026g); GLD reflects the price of gold bullion less trust expenses (State Street Global Advisors, 2026a); DBC is used as the broad-commodity sleeve (Invesco, 2026); USO is the dedicated crude-oil sleeve (USCF Investments, 2026); and VNQ captures listed U.S. real-estate exposure (Vanguard, 2026). The resulting panel separates assets whose stress behavior should plausibly differ: Treasuries capture duration and potential flight-to-quality exposure, high-yield credit represents risky corporate claims, gold and commodities capture real-asset exposure, oil isolates energy-specific risk, and listed real estate combines property exposure with equity-market trading characteristics.

Panel B: International equity universe. The international equity panel contains six ETFs:

$$\{\text{SPY}, \text{EEM}, \text{EWG}, \text{EWU}, \text{EWJ}, \text{EFA}\}.$$

This panel combines one U.S. benchmark ETF (SPY), one broad developed-markets ex-U.S. ETF (EFA), one broad emerging-markets ETF (EEM), and three country ETFs for Germany (EWG), the United Kingdom (EWU), and Japan (EWJ). The logic is to move from *cross-asset diversification* to *geographic equity diversification*. SPY remains the U.S. equity benchmark (State Street Global Advisors, 2026b); EFA tracks developed-market equities outside the United States and Canada (iShares, 2026c); EEM tracks emerging-market equities (iShares, 2026b); and EWG, EWU, and EWJ provide targeted exposure to German, U.K., and Japanese equities, respectively (iShares, 2026d,f,e). This allows the stress analysis to distinguish two related but conceptually different questions: whether international equities diversify U.S. equity exposure at all, and whether this geographic diversification survives once stress compresses global equity returns toward a common factor.

Methodologically, this second panel is useful because it preserves a common asset class (equities) while varying the geographic source of risk. If correlations rise in Panel B as sharply as in Panel A, the interpretation is not a generic “everything correlates in crises” result, but a more specific finding that *international equity diversification itself* becomes fragile in stress.

3.2 Sample Period and Data Sourcing

The empirical analysis is conducted at the *daily* frequency. Throughout the project, prices are sourced from Yahoo Finance via the `yfinance` Python interface. The package documents daily data downloads through the `interval='1d'` argument and provides automatic price adjustment functionality through the `auto_adjust` argument (Aroussi, 2026). In the project pipeline, Yahoo Finance is the source for ETF prices and VIX quotes, while the data are stored in the project database as daily adjusted-close observations (close levels for the VIX) before returns are constructed.

A crucial design choice is the sample window. Because the main tests compare assets within a fixed universe and estimate regime-conditional correlation matrices and PCA concentration, they require a *balanced* window in which the full universe is simultaneously observable: the balanced window is anchored at **11 April 2007**, the first trading day on which all ETFs are jointly available, with HYG as the latest-entering instrument (iShares, 2026g); the regime model itself is estimated on the full SPY history (Section 4). Estimating every regime-conditional matrix on the same trading days keeps the cross-regime comparison clean and, by avoiding pairwise deletion, yields an internally consistent positive-semidefinite matrix for the eigenvalue analysis (Lewis, 1995; Jolliffe, 2002). The descriptive tables, by contrast, report each ETF over its own longer availability window, so raw coverage is documented transparently while the main analysis prioritizes comparability and matrix coherence.

All return calculations are based on split- and distribution-adjusted daily prices obtained from Yahoo Finance via `yfinance`. This adjustment is important because unadjusted closing prices omit distributions and corporate actions, which would mechanically understate investor-relevant returns for payout-heavy instruments such as bond ETFs, high-yield credit ETFs, and REIT ETFs. The adjusted-price approach therefore approximates total-return exposure more closely than a price-only return series. Because Yahoo Finance and `yfinance` provide adjusted data fields and automatic adjustment functionality, the empirical return construction uses adjusted prices before computing returns (Aroussi, 2026).

3.3 Return Construction

Let $P_{i,t}^{adj}$ denote the split- and distribution-adjusted daily price of ETF i on trading day t . Daily ETF returns are computed as log returns:

$$r_{i,t} = \ln(P_{i,t}^{adj}) - \ln(P_{i,t-1}^{adj}). \quad (4)$$

The use of log returns is standard in financial econometrics and appropriate here for two reasons. First, log returns are *time-additive*: the multi-period log return over a block of consecutive trading days is the sum of the corresponding daily log returns. This is convenient for regime-based work because returns can be aggregated consistently from daily observations. Second, for small daily price changes, log returns are numerically close to simple percentage returns, while providing a convenient continuously compounded return representation used widely in empirical asset-pricing and time-series applications (Campbell et al., 1997; Tsay, 2010). Log returns are therefore used as the common return metric for all ETFs before the regime labels are merged into the return panel.

The ordering of the data construction is important. The daily return series is computed *after* price adjustment and *before* regime labeling. In other words, the project first builds a clean daily ETF return panel and only then merges in the stress labels constructed elsewhere in the pipeline. This ordering separates measurement choices in the price series from the later regime-classification procedure.

3.4 Descriptive Statistics and Data Availability

The descriptive statistics serve two purposes. First, they document coverage by reporting each ETF’s first available adjusted price observation and the corresponding number of daily observations. This is important because the main empirical analysis is conducted on a balanced panel starting only once all ETFs in the final universe are simultaneously observable. Second, they summarize the unconditional distribution of daily returns using annualized mean return, annualized volatility, skewness, and excess kurtosis. These moments are relevant for a stress-regime analysis because financial returns are well known to exhibit non-normal features such as asymmetry, volatility clustering, and fat tails (Fama, 1965; Cont, 2001). Reporting skewness and excess kurtosis therefore provides an initial diagnostic of the distributional properties that motivate the later regime-dependent analysis.

Appendix Table 7 reports the asset-level coverage in the raw adjusted-price dataset. Importantly, this table is *not* the balanced estimation window; it is a transparency table showing the underlying data availability by ETF. The descriptive moments are reported in Appendix Tables 8–9 as annualized mean return, annualized volatility, skewness, and excess kurtosis.

For the cross-asset panel, the moments show substantial heterogeneity in both average return and risk. SHY has by far the lowest annualized volatility, consistent with its role as a short-duration Treasury sleeve, whereas USO is the most volatile asset by a wide margin. AGG, HYG, and VNQ display particularly elevated excess kurtosis, indicating heavy

tails, and the pronounced negative skewness in the panel sits in AGG and USO. HYG’s full-sample skewness is, perhaps surprisingly, mildly positive: a handful of extreme crisis-rebound days (most notably +11.6% on 13 October 2008 and +6.3% on 9 April 2020) dominate the third moment, while outside those episodes high-yield returns are negatively skewed, consistent with the downside asymmetry of risky credit. Gold exhibits lower volatility than equities and commodities such as oil, but still materially non-Gaussian returns.

For the international equity panel, all six equity proxies are considerably more volatile than core bond proxies and non-U.S. equities retain clear tail risk. EEM has the highest annualized volatility in the panel, while the developed-market country ETFs and EFA cluster somewhat below that level. Across the board, excess kurtosis remains well above zero, again indicating that the later stress-regime analysis should not be built on Gaussian intuition alone.

Normal quantile–quantile plots for both panels are reported in Appendix B (Figures 6 and 7); under approximate normality the points would lie on the Gaussian reference line (Wilk and Gnanadesikan, 1968). Consistent with the skewness and excess-kurtosis columns of Appendix Tables 8–9, the plots keep their central mass near the benchmark but show pronounced S-shaped tail departures—most severely for AGG, HYG, USO, and VNQ—confirming the fat-tailed, asymmetric returns that motivate the regime-based analysis (Fama, 1965; Cont, 2001).

The descriptive statistics already point to strong cross-sectional heterogeneity in the return-generating process. Safe government-bond exposure (SHY) is characterized by low volatility, while oil (USO), listed real estate (VNQ), and emerging-market equities (EEM) are substantially more volatile. Several assets exhibit marked excess kurtosis and, in some cases, negative skewness, indicating that the relevant return distributions are asymmetric and heavy-tailed. These features motivate the later regime-based analysis of correlation structure under calm and stress states.

In summary, the data design combines an economically interpretable ETF universe with daily adjusted prices and log-return construction. The descriptive evidence confirms that the selected assets differ strongly in volatility and tail behavior, which makes them suitable for evaluating whether diversification benefits persist or erode once market stress regimes are imposed.

4 Methodology

This section defines the binary market state—*calm* versus *stress*—on which the rest of the analysis conditions. Because any single stress definition embeds discretionary choices, we use two complementary definitions and verify that they agree: a transparent, rule-based label built from the VIX and the S&P 500 drawdown, and a data-driven Markov-switching model estimated on S&P 500 returns. The rule-based label is easy to interpret but relies on fixed thresholds; the Markov-switching model instead lets the data themselves separate a low-volatility from a high-volatility regime. Throughout, the S&P 500 is proxied by the SPY ETF, using the split- and distribution-adjusted prices defined in Section 3.

4.1 Regime Definition

Baseline rule-based stress label. The first definition is a simple, transparent rule that flags a trading day as “stress” when two conditions hold simultaneously: implied volatility is unusually high, and the equity market is in a meaningful drawdown. Implied volatility is measured by the VIX, the CBOE index of S&P 500 option-implied volatility, which is widely interpreted as a forward-looking “fear gauge” of expected near-term market turbulence (Whaley, 2009). To avoid a fixed cut-off that would be sensitive to the secular level of volatility, we compare the VIX to its own recent history: a day is classified as elevated when the VIX exceeds the trailing 75th percentile of its previous 252 trading days (approximately one year). Let V_t denote the closing VIX level and $Q_t^{0.75}$ the 75th percentile of $\{V_{t-252}, \dots, V_{t-1}\}$.

The second condition requires a material equity drawdown. Let P_t^{SPY} be the adjusted SPY closing price and let $M_t = \max_{t-251 \leq s \leq t} P_s^{\text{SPY}}$ be its trailing one-year (252-day) running maximum. The drawdown is

$$D_t = \frac{M_t - P_t^{\text{SPY}}}{M_t}, \quad (5)$$

that is, the fractional decline from the recent peak. A day is labelled *stress* when both conditions hold:

$$\text{stress}_t = \mathbf{1}[V_t > Q_t^{0.75}] \cdot \mathbf{1}[D_t > 0.05]. \quad (6)$$

Requiring both an above-normal VIX *and* a drawdown exceeding 5% prevents the label from firing on volatility spikes that are not accompanied by falling prices, or on slow declines without elevated implied volatility. Over the sample this rule classifies roughly 16% of trading days as stress. As a robustness variant we also compute the same label after smoothing the VIX with a 21-day moving average, which removes single-day spikes; the two rule-based labels are very close. This label is deliberately simple and in-

interpretable, but the 75th-percentile and 5% thresholds are discretionary, which motivates the complementary model-based definition.

Official Markov-switching model. The second, and primary, regime definition replaces fixed thresholds with a statistical model that infers the regime directly from the return data. We use a two-state Markov-switching model in the tradition of Hamilton (1989), estimated on *monthly* SPY log returns. Monthly frequency is the standard choice for return-based regime models in the asset-allocation literature (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007): it averages out daily microstructure noise while leaving enough observations to estimate the regimes precisely (here $n = 311$ months, February 2000–December 2025).

Let r_t denote the monthly SPY log return (expressed in percent), and let $S_t \in \{0, 1\}$ be a latent state that follows a first-order Markov chain with transition probabilities

$$p_{ij} = \Pr(S_t = j \mid S_{t-1} = i), \quad i, j \in \{0, 1\}. \quad (7)$$

Conditional on the state, returns are Gaussian with a state-specific mean and variance:

$$r_t \mid (S_t = s) \sim \mathcal{N}(\mu_s, \sigma_s^2), \quad s \in \{0, 1\}. \quad (8)$$

Both the mean μ_s and the variance σ_s^2 are allowed to switch across states, and the parameters $\{\mu_0, \mu_1, \sigma_0^2, \sigma_1^2, p_{00}, p_{11}\}$ are estimated by maximum likelihood (Hamilton, 1989). To date the regimes we use the *smoothed* state probabilities $\Pr(S_t = s \mid r_1, \dots, r_T)$, which condition on the full sample and are the appropriate object for ex-post regime identification (Kim, 1994; Kim and Nelson, 1999); they describe the historical record rather than provide a real-time forecast.

The two states are not labelled a priori. We identify the *stress* state as the one with the higher fitted variance, following the standard interpretation of the high-volatility regime as the crisis state (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007). A period is then assigned to the stress regime using the textbook maximum-probability rule, i.e. when its smoothed stress probability is at least 0.5 (Hamilton, 1989; Kim and Nelson, 1999); we additionally retain a stricter 0.75 label as a high-confidence variant.

Figure 1 visualizes the fitted Markov chain that underlies this classification. Each month occupies one of the two latent states—a calm, low-volatility state and a stress, high-volatility state—and the transition probabilities govern whether the process persists in its current state or switches in the following month. Both fitted regimes are strongly persistent, with self-transition probabilities above 0.93 and implied average durations of roughly 16–17 months, so the model produces gradual, economically interpretable regime

shifts rather than frequent flickering. The stress state carries a monthly volatility more than twice that of the calm state together with a slightly negative average return, the pattern one expects of a crisis regime. Table 1 reports the same fitted state statistics numerically.

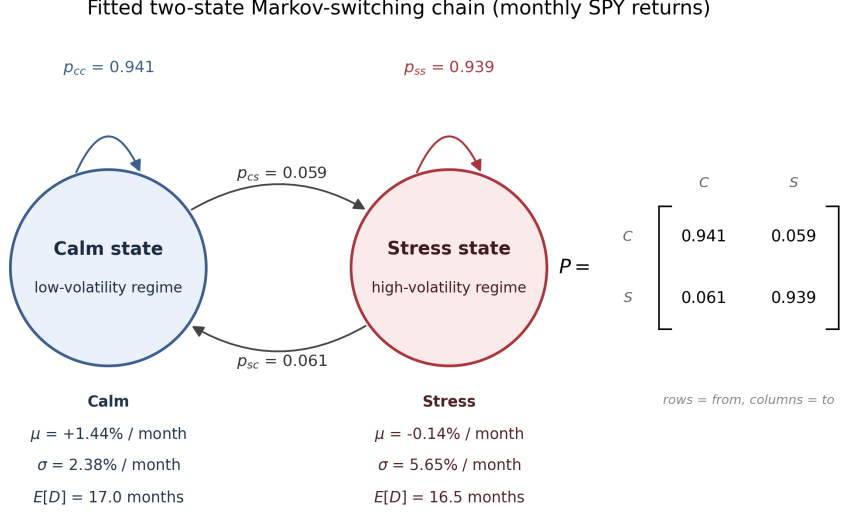


Figure 1: Fitted two-state monthly Markov-switching chain for SPY log returns (2000–2025). Each node is a latent state—a calm, low-volatility state and a stress, high-volatility state. The self-loops report the persistence (self-transition) probabilities p_{cc} and p_{ss} , and the connecting arrows the switching probabilities p_{cs} (calm-to-stress) and p_{sc} (stress-to-calm); the matrix P collects the four probabilities (rows: from state, columns: to state). Below each node are the state-specific mean μ and volatility σ (both in % per month) and the expected duration $E[D] = 1/(1 - p_{ii})$. Stress is identified ex post as the higher-variance state; the subscripts c and s denote the calm and stress states.

Table 1: Fitted two-state monthly Markov-switching model for SPY returns (2000–2025, $n = 311$). The stress state is the higher-variance state. Returns are monthly log returns in percent.

State	Mean (%/month)	Std. dev. (%/month)	Persistence p_{ii}	Expected duration (months)
Calm (state 0)	1.44	2.38	0.941	17.0
Stress (state 1)	−0.14	5.65	0.939	16.5

Log-likelihood = −870.6; AIC = 1753.2; BIC = 1775.6. Stress is identified as the higher-variance state.

The fitted model isolates every major episode in the sample—the dot-com unwind, the 2008 global financial crisis, the 2011 euro-area stress, the late-2018 selloff, the 2020 COVID-19 crash, and the 2022 rate-hike drawdown—without being told about any of them in advance. The full monthly regime path—SPY with the identified stress months shaded, the smoothed stress probability against the 0.5 and 0.75 thresholds, and a swim-lane comparison of the monthly Markov label with the rule-based and weekly Markov labels—is shown in Appendix Figure 8.

Specification choice. The choice of two states, and of switching *both* the mean and the variance, is not imposed but selected by information criteria. Table 10, reported in Appendix C, compares the official specification against natural alternatives. The two-state model that switches both the mean and the variance attains the lowest BIC (Schwarz, 1978), beating the variance-only and mean-only two-state specifications. The three-state models achieve a lower in-sample AIC, but they fail to converge to a stable optimum, and BIC—which penalizes additional parameters more heavily—selects the parsimonious two-state model. We therefore adopt the two-state, mean-and-variance-switching specification, which is both well identified and economically interpretable.

Robustness and agreement of the two definitions. We assess the regime definition along three dimensions. First, *threshold sensitivity*: tightening the probability cut-off from 0.5 to 0.75 reduces the share of months classified as stress from about 47% to 40%, without changing the timing of the episodes. Second, *sampling frequency*: re-estimating the identical model on weekly returns ($n = 1,356$ weeks, taken on the last trading day of each week) yields a sharper regime that flags about 26% of weeks as stress and isolates the same crises, confirming that the regime is not an artifact of monthly sampling (Figure 9, Appendix C). Third, and most importantly, *agreement between the two independent definitions*: we compare the model-based and rule-based stress labels on overlapping dates and summarize their agreement with Cohen’s κ (Cohen, 1960), which corrects the raw agreement rate for the fact that stress is rare—two labels can coincide simply by both calling most periods “calm”. Table 11, in the same appendix, reports the results. The rule-based and Markov labels agree well beyond chance, and the agreement is strongest at the weekly frequency ($\kappa \approx 0.52$, “moderate” on the scale of Landis and Koch (1977)), where the model’s stress share is closest to that of the rule. The weekly and monthly Markov labels themselves agree at $\kappa \approx 0.49$. That two methodologically unrelated definitions—one a fixed rule on the VIX and drawdowns, the other an unsupervised model on returns—independently identify the same stress episodes is the central validation of the regime definition used in the remainder of the paper.

Estimation robustness. A model-based regime is only credible if its fitted parameters represent a genuine likelihood optimum rather than an artifact of the optimizer’s starting values; because the Markov-switching likelihood is multimodal, this is a real concern. We therefore re-estimate the official specification with a substantially more aggressive initialization—additional EM warm-up iterations together with twenty random starting-value searches (`em_iter=20`, `search_reps=20`)—and compare it against the default fit. Table 12, in Appendix C, shows the two fits to be numerically indistinguishable: the log-likelihood, information criteria, state variances, transition probabilities, and stress

share are unchanged to three significant figures, and the smoothed stress-probability paths correlate at $\rho \approx 1.00$, assigning the identical 0.5 stress label in every month of the sample. The regime dating is therefore not an artifact of estimation initialization.

In summary, the analysis conditions on a stress regime that is defined in two complementary ways and validated by their agreement. The official label is the monthly two-state Markov-switching model, with stress identified as the high-variance state and assigned by the maximum-probability rule; the rule-based VIX-plus-drawdown label and the weekly Markov model serve as transparent and frequency-based robustness checks.

4.2 Regime-Conditional Correlations

The first and most direct question is whether assets that appear to diversify one another in calm markets continue to do so under stress. We answer it with regime-conditional pairwise correlations, which require no distributional assumptions beyond the existence of second moments: they summarize the comovement a portfolio holder actually experienced in each state. Each trading day t inherits the binary stress label of its month from the monthly Markov-switching model of Section 4.1; let $m_t \in \{0, 1\}$ denote this daily indicator, with $m_t = 1$ in stress and $m_t = 0$ in calm.

For each unique unordered pair of assets (i, j) with $i < j$ we estimate the Pearson correlation of daily log returns separately within each regime,

$$\rho_{ij}^{\text{calm}} = \text{corr}(r_{i,t}, r_{j,t} \mid m_t = 0), \quad \rho_{ij}^{\text{stress}} = \text{corr}(r_{i,t}, r_{j,t} \mid m_t = 1), \quad (9)$$

where $r_{i,t}$ is the daily log return of Equation (4). The central object of interest is the change in correlation across regimes,

$$\Delta\rho_{ij} = \rho_{ij}^{\text{stress}} - \rho_{ij}^{\text{calm}}, \quad (10)$$

which we refer to as the *breakdown* of a pair: a large positive $\Delta\rho_{ij}$ means the two assets move together far more tightly in stress than in calm, eroding exactly the diversification that justified holding them jointly. Ranking pairs by $\Delta\rho_{ij}$ isolates the relationships that fail when diversification is most needed.

To summarize dependence at the portfolio level rather than pair by pair, we report, for each regime $g \in \{\text{calm}, \text{stress}\}$, the average, median, and the 10th and 90th percentiles of the set of unique pairwise correlations $\{\rho_{ij}^g\}$ across the $N(N-1)/2$ pairs of the panel. Comparing these distributions between regimes shows whether the whole dependence structure shifts upward in stress or whether the move is concentrated in a few pairs.

A measurement caveat governs the sample used for every quantity. Portfolio-level aggregates—

the average pairwise correlation, and the concentration measures of Section 4.5—are computed on the *balanced* return panel that starts on 12 April 2007, one trading day after the last-entering asset’s first price observation on 11 April (Section 3.2). Every asset therefore contributes the same return dates and the resulting correlation matrix is internally consistent and positive semidefinite, which matters for the later eigenvalue analysis (Lewis, 1995; Jolliffe, 2002). The international-equity panel (Panel B) contains no asset with a 2007 inception and could in principle begin earlier; we hold it to the same common window so the cross-panel comparison is on identical days, and verify in Appendix F that its full 2003–2025 sample delivers the same concentration verdict. Pair-level quantities in Equations (9)–(10) are deliberately estimated on this same shared calendar rather than on each pair’s maximal available history. They are therefore directly comparable across panels, but conditional on the availability window of the combined universe. These raw correlations are the benchmark against which the variance-adjusted measure of Section 4.4 is compared.

4.3 Bootstrap Inference for Correlation Changes

The regime-conditional changes $\Delta\rho_{ij}$ of Equation (10) are point estimates; deciding whether a given change is distinguishable from zero requires its sampling distribution. Two features of the data make textbook inference for a correlation difference inappropriate. First, daily returns are serially dependent—volatility clusters, so adjacent days are not independent draws (Cont, 2001)—which violates the i.i.d. assumption behind the classical Fisher- z test. Second, the calm and stress samples are not two independent random samples but the regime-sorted subsequences of a single time series. We therefore base inference on a *moving-block bootstrap*, the standard resampling scheme for dependent data (Efron, 1979; Künsch, 1989; Lahiri, 2003).

For each universe we resample the full chronologically ordered daily panel using a *circular* moving-block bootstrap (Politis and Romano, 1992). Block starting points are drawn uniformly and blocks of L genuinely consecutive trading-day rows are concatenated, wrapping only at the end of the complete sample. Each row carries both the full return vector and its observed regime label. We split a replicate into calm and stress only after resampling, so blocks preserve real temporal adjacency and regime durations while the calm/stress counts are allowed to vary across replicates. Crucially, *whole cross-sectional rows* are resampled together, preserving contemporaneous dependence among all assets. We set the block length to $L = 21$ trading days—roughly one trading month—to retain short-run autocorrelation and volatility clustering, and assess sensitivity to $L \in \{5, 21, 63\}$ in Appendix E. Each of $B = 2,000$ replicates b yields a full set of resampled changes $\Delta\rho_{ij}^{(b)}$.

We summarize the bootstrap distribution of each $\Delta\rho_{ij}$ by a percentile 95% confidence interval—its 2.5th and 97.5th sample quantiles—and use the centered bootstrap errors for an approximate two-sided test of $H_0 : \Delta\rho_{ij} = 0$,

$$p_{ij} = \frac{1 + \# \left\{ b : \left| \Delta\rho_{ij}^{(b)} - \widehat{\Delta\rho}_{ij} \right| \geq \left| \widehat{\Delta\rho}_{ij} \right| \right\}}{B + 1}, \quad (11)$$

with the $(\cdot + 1)/(B + 1)$ continuity correction of Davison and Hinkley (1997); the smallest attainable value is therefore $1/(B + 1) \approx 0.0005$. A pair is a *significant increase (decrease)* when its 95% interval lies entirely above (below) zero. Because each universe tests many pairs at once—36, 15, and 91 in Panel A, Panel B, and the combined universe—we control the false discovery rate within each universe at $q < 0.05$ using the Benjamini and Hochberg (1995) procedure, and report FDR-significant counts alongside the raw 95% counts.

As a deliberately permissive benchmark we also report the classical Fisher- z two-sample correlation test (Fisher, 1921), which treats the two regime correlations as independent estimates from i.i.d. Gaussian samples. The contrast is itself informative: where the block bootstrap is markedly more conservative than Fisher- z , the gap measures the over-rejection induced by ignoring serial dependence.

One limitation remains important. Regime labels travel with the resampled rows, so the bootstrap preserves their observed local timing but treats the estimated Markov classification of Section 4.1 as data. It does not refit the Markov model inside every replicate and therefore does not propagate model or classification uncertainty. The reported intervals remain conditional on the fitted regime definition.

4.4 Forbes–Rigobon Variance Adjustment

The regime-conditional correlations of Section 4.2 answer what a holder experienced, but they cannot by themselves say *why* correlations rose. Forbes and Rigobon (2002) show that a measured correlation mechanically increases when the variance of the common driving factor increases, even if the underlying linear relationship between two assets is unchanged. Because stress periods are by construction high-variance, part—or all—of the observed correlation rise can be this volatility bias rather than a genuine tightening of dependence. The concern is acute in our design because the stress regime is itself identified by high volatility: Boyer et al. (1999) show that correlations estimated on subsamples sorted on volatility differ mechanically from the full-sample value even when the underlying dependence is constant, so regime-conditional correlations are precisely the setting in which this correction is required. The Forbes–Rigobon adjustment therefore answers

a distinct question from Section 4.2: did the dependence relationship itself change, or is the stress correlation rise a mechanical consequence of inflated variance? This distinction concerns the *source* of the breakdown, not whether it occurs. The raw correlations remain the comovement an investor actually holds in either case, so a volatility-driven verdict identifies *interdependence*—heightened comovement in a more volatile state—rather than *contagion*—a structural shift in the dependence relationship—without implying that realized diversification was preserved. The adjustment thus diagnoses the mechanism behind the experienced breakdown; it does not undo it.

We implement the adjustment through a per-asset variance shock. For asset i , let

$$\phi_i = \frac{\text{Var}(r_{i,t} \mid m_t = 1)}{\text{Var}(r_{i,t} \mid m_t = 0)} \quad (12)$$

be the ratio of stress to calm return variance, and define the (non-negative) variance shock

$$\delta_i = \max(\phi_i - 1, 0). \quad (13)$$

Following Forbes and Rigobon (2002), the stress correlation of a pair is then deflated for the excess variance carried by the common factor,

$$\tilde{\rho}_{ij}^{\text{stress}} = \frac{\rho_{ij}^{\text{stress}}}{\sqrt{1 + \delta_{ij} \left(1 - \left(\rho_{ij}^{\text{stress}}\right)^2\right)}}, \quad (14)$$

where δ_{ij} is the variance shock attributed to the pair (specified by the variant below). The adjusted breakdown is the analogue of Equation (10) using the corrected stress correlation,

$$\tilde{\Delta}\rho_{ij} = \tilde{\rho}_{ij}^{\text{stress}} - \rho_{ij}^{\text{calm}}. \quad (15)$$

The correction is not free. Equation (14) is exact only under the Forbes–Rigobon assumptions: returns are driven by a single common factor, and the regime shift changes *only* the variance of that factor, with no contemporaneous shift in fundamentals, in the factor loadings, or in the variance of the idiosyncratic shocks. If those structural objects also move between calm and stress, the adjustment over- or under-corrects. We state this assumption explicitly because it is the price the Forbes–Rigobon measure pays for separating genuine contagion from mechanical variance inflation, and naming it is what keeps the comparison with the raw correlations honest.

We report two variants that differ only in how the pair-level shock δ_{ij} in Equation (14) is chosen:

- **market_spy** (main): $\delta_{ij} = \delta_{\text{SPY}}$ for every pair, using SPY’s stress-versus-calm

variance shock as the single common market shock. This is the specification most faithful to the single-factor logic of Forbes and Rigobon (2002) and is the one carried in the body.

- **pair_max** (conservative): $\delta_{ij} = \max(\delta_i, \delta_j)$, the larger of the two assets' own variance shocks. By deflating with the bigger shock it removes the most volatility bias and is the most demanding robustness check. Treating an asset's own variance shock as the common-factor shock is a deliberately conservative heuristic extension of the single-factor logic of Forbes and Rigobon (2002), not the classical adjustment; this is why **market_spy** is the main specification and **pair_max** only a stress test.

Because the adjustment attributes the shock to a single designated source market, which asset plays that role is in principle directional; Forbes and Rigobon (2002) fix it by naming the crisis-origin market, and we make SPY that source throughout, so we do not report a separate per-asset (*directed*) split.

Combining the raw breakdown of Equation (10) with the adjusted breakdown of Equation (15) classifies each pair into the four-class fragility taxonomy defined in Section 4.6.

The Forbes–Rigobon adjustment is treated as a primary result rather than a caveat: as the Results section shows, it materially reshapes the headline, because much of the apparently dramatic stress-period breakdown turns out to be volatility-driven rather than a genuine change in dependence. Presenting the raw and adjusted measures side by side is therefore central to the paper's contribution, not a footnote to it.

4.5 Principal-Component Concentration

The pairwise measures of Sections 4.2–4.4 describe how individual relationships behave; they do not say how many genuinely independent sources of risk a portfolio retains. We answer that with a principal-component analysis of the regime-conditional correlation matrix, following Jolliffe (2002). The eigendecomposition is performed on the *raw* correlation matrix of Equation (9), not on the Forbes–Rigobon–adjusted matrix of Section 4.4; the concentration measures therefore describe the dependence structure as realized, consistent with the raw correlations.

Let R^g denote the $N \times N$ correlation matrix estimated within regime $g \in \{\text{full, calm, stress}\}$ on the balanced panel, with eigenvalues $\lambda_1^g \geq \lambda_2^g \geq \dots \geq \lambda_N^g \geq 0$. Because R^g is a correlation matrix, $\sum_{k=1}^N \lambda_k^g = N$. The share of total variance captured by the leading component and by the leading k components are

$$\text{pc1_share}^g = \frac{\lambda_1^g}{N}, \quad \text{top}k_share^g = \frac{1}{N} \sum_{l=1}^k \lambda_l^g, \quad k \in \{3, 5\}. \quad (16)$$

A high `pc1_share` means a single common direction—typically a market factor—explains most of the joint variation, leaving little room for independent diversifying exposures.

To make “how many independent bets” precise we use the effective number of bets, defined through the participation ratio of the eigenvalue spectrum (Meucci, 2009),

$$N_{\text{eff}}^g = \frac{\left(\sum_{k=1}^N \lambda_k^g\right)^2}{\sum_{k=1}^N \left(\lambda_k^g\right)^2} = \frac{N^2}{\sum_{k=1}^N \left(\lambda_k^g\right)^2}. \quad (17)$$

N_{eff}^g equals N when all eigenvalues are equal (every direction contributes equally, maximal diversification) and falls toward 1 as the spectrum concentrates into a single dominant component. We summarize the regime comparison with a single verdict: a panel is *more concentrated in stress* when $\text{pc1_share}^{\text{stress}} > \text{pc1_share}^{\text{calm}}$ and $N_{\text{eff}}^{\text{stress}} < N_{\text{eff}}^{\text{calm}}$, i.e. the leading factor grows and the effective number of bets shrinks at the same time.

A rising `pc1_share` together with a falling N_{eff} means that, precisely when stress arrives, the portfolio collapses onto fewer independent return directions, so its realized diversification weakens even if no single pairwise correlation looks alarming. This portfolio-level concentration is the bridge to the fragility taxonomy of Section 4.6.

4.6 A Fragility Taxonomy

The preceding measures combine into a simple taxonomy of how a relationship behaves under stress. Each pair is assigned to exactly one class from the joint behavior of its raw breakdown $\Delta\rho_{ij}$ (Equation (10)) and its variance-adjusted breakdown $\tilde{\Delta}\rho_{ij}$ (Equation (15)), on a fixed breakdown threshold of 0.10:

Robust breakdown. $\Delta\rho_{ij} \geq 0.10$ and $\tilde{\Delta}\rho_{ij} \geq 0.10$. The correlation rise survives the variance adjustment: a genuine loss of diversification in stress, the most fragile class.

Volatility-bias-sensitive. $\Delta\rho_{ij} \geq 0.10$ but $\tilde{\Delta}\rho_{ij} < 0.10$. The apparent breakdown is largely mechanical volatility inflation; the dependence relationship is more stable than the raw correlation suggests.

Mostly resilient. $0 \leq \Delta\rho_{ij} < 0.10$. A small, non-negative raw rise below the threshold, with no material breakdown either before or after adjustment.

Resilient (decreased). $\Delta\rho_{ij} < 0$. The pair diversifies *better* in stress, the opposite of fragility.

The four classes partition the pairs by raw breakdown—negative, below the threshold, or at least 0.10—with the last band split by whether the rise survives adjustment, so every pair falls in exactly one class. Read together with the portfolio-level concentration of Section 4.5, this taxonomy distinguishes diversification that is genuinely lost in stress from diversification that only *appears* to be lost because variance has risen—the central distinction the paper is built to make.

5 Results

This section presents the empirical results in the order the methodology builds them. We begin with the regime-conditional correlations themselves—the comovement a portfolio holder actually experienced in each state (Section 5.1)—and then establish which of the observed changes survive sampling uncertainty using the block bootstrap of Section 4.3 (Section 5.2). We move from pairs to portfolios with the principal-component concentration (Section 5.3) and finally ask whether the raw increases survive the Forbes–Rigobon variance adjustment (Section 5.4).

5.1 Regime-Conditional Correlations

The regime-conditional correlations already tell two different stories about the two universes, and a third—more subtle—about their combination. Table 2 reports the panel-level summaries on the common balanced window, and Figure 2 shows the full distribution of the pairwise stress-minus-calm change $\Delta\rho_{ij}$ of Equation (10).

Table 2: Regime-conditional correlation summary by universe, on the common balanced window. $\bar{\rho}^{\text{calm}}$ and $\bar{\rho}^{\text{stress}}$ are the mean pairwise Pearson correlations (Equation (9)) in each regime; $\overline{\Delta\rho}$ and the median summarize the stress-minus-calm change (Equation (10)). The last three columns give the share of pairs that are *resilient* ($\Delta\rho < 0$), a *small increase* ($0 \leq \Delta\rho < 0.10$), or a raw *breakdown* ($\Delta\rho \geq 0.10$).

Universe	N	Pairs	Mean ρ		$\Delta\rho$		Pair shares		
			Calm	Stress	Mean	Median	Resilient	Small $\uparrow$	Breakdown
Panel A: cross-asset	9	36	0.195	0.170	−0.025	−0.024	61.1%	25.0%	13.9%
Panel B: international equity	6	15	0.741	0.866	+0.124	+0.125	0.0%	26.7%	73.3%
Combined ETF universe	14	91	0.282	0.311	+0.029	+0.050	41.8%	24.2%	34.1%

Common balanced return window 12 April 2007–30 December 2025: 2,443 calm and 2,268 stress trading days (48.1% stress). The breakdown class ($\Delta\rho \geq 0.10$) is later split into robust and volatility-bias-sensitive by the Forbes–Rigobon adjustment (Section 5.4); the per-universe class counts are reported in Appendix Table 14.

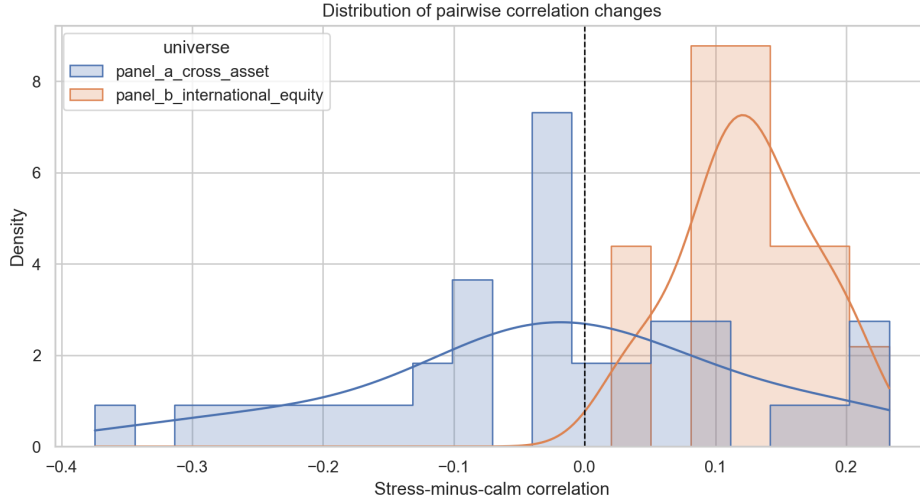


Figure 2: Distribution of pairwise stress-minus-calm correlation changes $\Delta\rho_{ij}$ (Equation (10)), with kernel-density overlays for the cross-asset (Panel A) and international-equity (Panel B) universes; each observation is one asset pair and the dashed vertical line marks $\Delta\rho = 0$. Panel B lies entirely to the right of zero—every pair tightens in stress—whereas Panel A is centred slightly to the *left* of zero, reflecting the reshuffle in which risk-on pairs tighten while safe-haven pairs decouple.

Panel B: international equity. The international-equity panel shifts upward as a whole. The average pairwise correlation rises from 0.74 in calm to 0.87 in stress, and *every one of the fifteen pairs* co-moves more tightly, so the distribution in Figure 2 lies entirely to the right of zero. The tightening is led by the Japan pairs—EEM–EWJ (+0.21), EWU–EWJ (+0.19), and EWG–EWJ (+0.18)—while the already near-collinear developed-market pairs EFA–EWG and EFA–EWU, both around 0.91 in calm, have little room left to rise (about +0.04). Nearly three-quarters of the pairs cross the 0.10 raw-breakdown threshold and none falls.

Panel A: cross-asset. The cross-asset panel is the more revealing case, because its average barely moves—if anything it falls, from 0.19 to 0.17 ($\overline{\Delta\rho} = -0.025$)—yet this flat average conceals a reshuffle. The risk-on complex tightens: SPY–VNQ rises by 0.21, DBC–VNQ and USO–VNQ by about 0.23, and SPY–USO and SPY–DBC by about 0.17. At the same time the safe-haven sleeve decouples further, and the largest moves in the whole panel are *decreases*: TLT–VNQ falls by 0.37, SHY–VNQ by 0.31, AGG–VNQ by 0.28, and the bond–bond pairs TLT–AGG and SHY–AGG by about 0.23 and 0.21 as duration and credit risk separate. Consistent with this split, 61.1% of the thirty-six pairs see their correlation *fall* in stress (the *resilient (decreased)* class of Section 4.6), only one pair in seven crosses the 0.10 breakdown threshold, and the distribution in Figure 2 is centred slightly to the left of zero—the opposite of the textbook breakdown.

The combined fourteen-asset universe sits between the two: the average correlation rises

modestly from 0.28 to 0.31 ($\overline{\Delta\rho} = +0.029$), and the distribution is near-symmetric about zero, with 34.1% of the ninety-one pairs above the breakdown threshold and 41.8% falling. Its largest increases mix the real-estate-equity and real-estate-commodity pairs that bridge the two panels—VNQ-EWU (+0.23), DBC-VNQ and USO-VNQ (about +0.23), and VNQ-EEM and VNQ-EWG (about +0.22)—while its largest decreases are exactly the cross-asset hedges of Panel A (TLT-VNQ -0.37 , SHY-VNQ -0.31). The full pair rankings for all three universes, together with the regime-conditional heatmaps and calm-versus-stress scatter plots, are reported in Appendix D (Tables 13–14 and Figures 10–11).

A descriptive stress-episode split adds one qualification to these pooled estimates. Appendix Table 15 recomputes selected hedge correlations on Markov-stress days only, separately for the 2007–2019, 2020–2021, and 2022–2023 episodes within the same balanced window. The hedge correlations are not constant across stress types: SPY-TLT is strongly negative in the earlier episodes (-0.48 in 2007–2019 stress, -0.45 in 2020–2021 stress) but turns mildly positive ($+0.10$) in the inflation- and rate-driven 2022–2023 episode, and the same reversal runs through the duration sleeve (TLT-VNQ moves from -0.35 to $+0.27$; TLT-AGG rises from 0.51 to 0.91). Gold is the exception: SPY-GLD stays within $[+0.01, +0.15]$ in every episode. Because the 2007–2019 and 2020–2021 episodes supply roughly three-quarters of the stress days, the pooled Treasury-hedge estimate is dominated by these earlier, deflationary-type episodes—consistent with the macro-finance evidence that the sign of stock–bond comovement depends on whether real or inflationary shocks drive the downturn (Campbell et al., 2020). This does not overturn the pooled Panel A result; it clarifies its interpretation: the Markov label identifies *when* markets are stressed, not *which* macroeconomic shock generates the stress, so the resilience of sovereign duration is best read as an average stress-regime property rather than an unconditional hedge guarantee. The split is descriptive and is not used for formal inference.

These raw labels are descriptive thresholds on the point estimates; before reading economic content into any single pair, we ask which of the changes survive sampling uncertainty.

5.2 Statistical Significance of the Correlation Changes

To separate signal from sampling noise we apply the circular block bootstrap of Section 4.3 ($B = 2,000$ replicates, block length $L = 21$ days) to every pairwise change and to the panel averages. Table 3 reports the universe-level results.

The panel averages confirm only one of the three stories outright. The international-equity average is the sole universe whose mean change is unambiguously positive: $\overline{\Delta\rho} =$

Table 3: Block-bootstrap significance of the regime-conditional correlation changes. $\overline{\Delta\rho}$ is the average stress-minus-calm pairwise change; the 95% confidence interval and two-sided p -value (Equation (11)) come from $B = 2,000$ circular moving-block bootstrap replicates with block length $L = 21$ days. “Sig. $\uparrow$ ” and “Sig. $\downarrow$ ” count pairs whose 95% interval lies entirely above or below zero; “FDR” counts pairs significant after the per-universe Benjamini–Hochberg correction at $q < 0.05$.

Universe	Pairs	$\overline{\Delta\rho}$	95% CI	p_{boot}	Sig. $\uparrow$	Sig. $\downarrow$	FDR
Panel A: cross-asset	36	−0.025	[−0.066, +0.019]	0.267	6	8	11
Panel B: international equity	15	+0.124	[+0.084, +0.157]	< 0.001	15	0	15
Combined ETF universe	91	+0.029	[−0.012, +0.068]	0.158	35	16	50

The bootstrap resamples chronological blocks of whole cross-sectional rows together with their regime labels (random seed 20250624); significance is conditional on the estimated Markov labels (Section 4.3).

The smallest attainable centered-bootstrap p -value is $1/(B + 1) \approx 0.0005$.

+0.124, with a 95% confidence interval of $[+0.084, +0.157]$ that excludes zero, a bootstrap p -value at the resolution floor ($p < 0.001$), and *all fifteen pairs* significant after the Benjamini–Hochberg correction. By contrast, the cross-asset and combined averages are *not* distinguishable from zero: their intervals straddle zero ($[-0.066, +0.019]$ and $[-0.012, +0.068]$) with bootstrap p -values of 0.27 and 0.16.

A null average, however, is not the absence of an effect—it is the cancellation of offsetting ones. The cross-asset panel illustrates this directly: although its *average* change is indistinguishable from zero, six pairs are significant increases and eight are significant decreases at the 95% level, and eleven survive the FDR correction. The significant increases are the risk-on real-estate pairs (SPY–VNQ, DBC–VNQ, USO–VNQ) and the significant decreases are the safe-haven pairs (TLT–VNQ, SHY–VNQ, AGG–VNQ, TLT–AGG, SHY–AGG); the panel average is flat precisely because these move in opposite directions. The reshuffle described in Section 5.1 is therefore statistically real even though the headline average is not. In the combined universe the same logic produces 35 significant increases against 16 decreases, 50 of which survive the FDR control.

These counts are markedly smaller than a naive analysis would suggest, and the gap is itself a result. The classical Fisher- z test, which assumes i.i.d. Gaussian returns, flags 73 of the 91 combined pairs as significant; the block bootstrap, which respects the serial dependence of daily returns, retains only 50 after FDR control (Appendix Table 18). Every pair the bootstrap declares significant is also significant under Fisher- z , but 23 pairs are significant under Fisher- z alone—almost a third of its discoveries overturned by the more honest variance estimate. One pair makes the point concrete: SPY–AGG has $\Delta\rho = +0.08$ with a Fisher- z p -value of 0.008, yet a bootstrap 95% interval of $[-0.076, +0.201]$ and $p = 0.27$. The classical test treats 4,711 serially correlated, heteroskedastic days as independent observations and so overstates the effective sample size; the block bootstrap does not, and we read its more conservative verdict as the credible one.

Three caveats temper even this conservative reading, and all point the same way—towards *wider* uncertainty than reported. First, the bootstrap is conditional on the estimated Markov regime labels: it does not propagate the uncertainty in *which* days are stress, so a pair’s interval understates total uncertainty whenever the regime classification is itself in doubt. Second, the per-universe FDR families overlap—the combined universe reuses every pair from Panels A and B—so the significance counts across the three universes are not independent discoveries. Third, with $B = 2,000$ the centered-bootstrap p -value cannot fall below $1/(B + 1) \approx 0.0005$, so values at that floor are right-censored: the procedure cannot rank degrees of significance among the most extreme pairs. None of these changes the qualitative conclusion, but each argues against over-interpreting a borderline pair.

The conclusions are not an artifact of the 21-day block length. Re-running the bootstrap with $L \in \{5, 21, 63\}$ leaves the point estimates unchanged and keeps the international-equity average significant at every block length, while the confidence intervals widen as the block lengthens—the expected consequence of acknowledging more serial dependence (Appendix E, Figure 14 and Table 17). The pair-level bootstrap distributions and the full set of pairwise intervals for the top and tail pairs are reported there as well (Figures 12–13 and Table 16). Having established which correlations move and which of those moves are statistically real, we now ask what they imply for a portfolio as a whole.

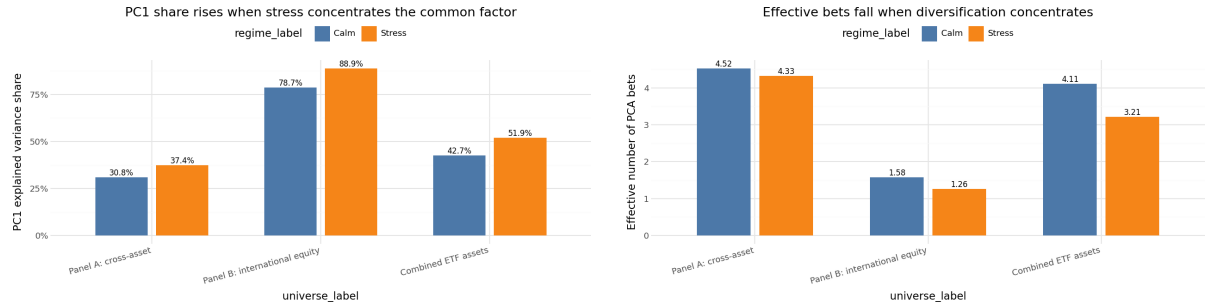
5.3 Principal-Component Concentration

We summarize each panel by the share of variance captured by the leading component, pc1_share (Equation (16)), and the effective number of bets, N_{eff} (Equation (17)). A panel is judged *more concentrated in stress* when the leading factor grows and the effective number of bets shrinks at the same time. Table 4 and Figure 3 report the comparison, and the verdict is unanimous: in all three universes pc1_share rises and N_{eff} falls from calm to stress. The two measures are different summaries of the same eigenvalue spectrum and move together almost by construction, so they are best read as one concentration finding rather than two independent confirmations; what the comparison establishes is its sign and that it holds in every universe. A rising leading-component share is precisely the signature that the systemic-risk literature reads as fragility—the “absorption ratio” of Kritzman et al. (2011), who show that a larger fraction of total variance loading on a few components coincides with market turbulence and predicts drawdowns, and the principal-component connectedness of Billio et al. (2012), which tightens in crises. Our regime-conditional estimates make the same point in the cross-section: precisely when stress arrives, returns collapse onto fewer independent directions, so the realized diversification of a portfolio weakens even where no single pairwise correlation looks alarming.

Table 4: Principal-component concentration by regime. $pc1_share$ is the share of total variance explained by the leading eigenvector of the regime-conditional correlation matrix (Equation (16)); N_{eff} is the effective number of bets (Equation (17)), which ranges from 1 (a single common factor) to N (equal, independent directions). Every panel is *more concentrated in stress*: the leading factor grows and the effective number of bets shrinks simultaneously.

Universe	N	$pc1_share$			N_{eff}			Verdict
		Calm	Stress	Δ	Calm	Stress	Δ	
Panel A: cross-asset	9	0.308	0.374	+0.065	4.52	4.33	-0.20	concentrates
Panel B: international equity	6	0.787	0.889	+0.101	1.58	1.26	-0.32	concentrates
Combined ETF universe	14	0.427	0.519	+0.092	4.11	3.21	-0.90	concentrates

Eigendecomposition of the raw (unadjusted) regime-conditional correlation matrix of Equation (9), computed on the common balanced return panel (12 April 2007 onward, one trading day after the last asset’s first price), so all three universes share the same 2,443 calm / 2,268 stress trading days (48.1% stress). Panel B’s full 2003–2025 sample yields the same verdict and is reported in Appendix Table 19.



(a) Leading-component share $pc1_share$.

(b) Effective number of bets N_{eff} .

Figure 3: Portfolio concentration by regime. In every universe the leading component absorbs a larger share of variance in stress (panel a) while the effective number of independent bets falls (panel b)—the two halves of the concentration verdict. The joint $(\Delta pc1_share, \Delta N_{eff})$ scatter is reported in Figure 16.

Panel A: cross-asset. The cross-asset universe is the more revealing case, because its diversification erodes at the portfolio level even though its average pairwise correlation does not rise. The leading component climbs from $\text{pc1_share}^{\text{calm}} = 0.308$ to $\text{pc1_share}^{\text{stress}} = 0.374$, while the effective number of bets falls from $N_{\text{eff}}^{\text{calm}} = 4.52$ to $N_{\text{eff}}^{\text{stress}} = 4.33$. In calm the panel is organized around two comparable factors— pc1_share and the second-component share are almost equal (0.308 versus 0.290)—which is the eigenvalue signature of a portfolio that genuinely runs a risk axis and a separate rates axis side by side. In stress the leading axis pulls away (0.374 versus a second-component share of 0.239), yet the top-three and top-five cumulative shares are essentially unchanged (≈ 0.76 and ≈ 0.89): the concentration is a reallocation of variance *from* the second component *into* the first, not an increase in the total explained by the leading components. This is exactly what the pairwise reshuffle implies. The risk-on block converges onto a single direction—SPY–VNQ, SPY–DBC, and SPY–USO all rise by 0.17–0.21, and the commodity–real-estate pairs DBC–VNQ and USO–VNQ rise by about 0.23—while the safe block moves more sharply against it, with SPY–TLT, TLT–AGG, and TLT–VNQ all falling (the *resilient (decreased)* class of Section 4.6). Both movements load onto the same leading risk-on/risk-off eigenvector, which is why pc1_share rises while the average correlation stays flat. The practical reading is that the cross-asset portfolio retains its hedges—Treasuries and gold still diversify, and indeed diversify better in stress—but loses roughly one-fifth of an independent bet as the remaining risk assets begin to move as one.

Panel B: international equity. The international-equity panel shows the textbook breakdown at the portfolio level, matching its pairwise behavior. It is already close to a single-factor system in calm, with $\text{pc1_share}^{\text{calm}} = 0.787$ and only $N_{\text{eff}}^{\text{calm}} = 1.58$ effective bets across six markets, and it becomes nearly degenerate in stress: $\text{pc1_share}^{\text{stress}} = 0.889$ and $N_{\text{eff}}^{\text{stress}} = 1.26$, barely more than one independent direction, with the second-component share roughly halving from 0.077 to 0.039. This is the eigenvalue counterpart of a uniform pairwise tightening in which all fifteen pairs rise: the increase is led by the Japan pairs (EEM–EWJ by 0.21, EWU–EWJ by 0.19, EWG–EWJ by 0.18, SPY–EWJ by 0.16), while the already tight developed-market pairs EFA–EWG and EFA–EWU remain near 0.95 (rising by only about 0.04). The common global-equity factor that dominates in calm becomes very nearly the only factor in stress, so the diversification benefit of holding geographically distinct equity markets—the variance *not* explained by the common component—all but disappears. This is the cross-sectional, regime-conditional analogue of the rising global-market integration documented by Pukthuanthong and Roll (2009), and of the extreme-downturn correlations of Longin and Solnik (2001): international equity diversification is weakest exactly in the states where it is most needed. The verdict

is unchanged on Panel B’s full 2003–2025 sample (Appendix Table 19), which the body shortens to the common window only for comparability with the other panels.

The combined fourteen-asset universe sits between the two by construction, but records the largest absolute loss of independent bets: `pc1_share` rises from 0.427 to 0.519 and N_{eff} falls from 4.11 to 3.21, a drop of nearly a full bet, as the international-equity block drags the leading component upward. Across all three universes the joint verdict holds—`pc1_share` up and N_{eff} down (Figure 16)—so the portfolio-level concentration is not an artifact of a single asset mix. Read together with the pairwise fragility taxonomy of Section 4.6, this establishes that the realized loss of diversification in stress operates at the level of the whole raw correlation matrix. The Forbes–Rigobon adjustment below asks whether the raw pairwise increases that feed this concentration survive a stricter variance correction.

5.4 Forbes–Rigobon Adjustment

The PCA results describe the realized stress-period concentration of the full correlation matrix. The Forbes–Rigobon adjustment asks a deliberately narrower question. It does *not* dispute that diversification weakens in stress—the raw correlations are the comovement an investor actually holds, and they rise—but asks *why*: did the dependence relationship itself change (*contagion*: a structural shift in factor loadings or genuinely new linkages), or did the measured correlation rise mechanically because the common factor became more volatile while the underlying relationship was unchanged (*interdependence*)? Forbes and Rigobon (2002) show that the second channel alone inflates a correlation even when the linear relation is fixed. This warning is especially relevant here because the regimes are deliberately sorted by volatility: Boyer et al. (1999) show that volatility-conditioned subsamples can display different correlations even when the underlying dependence is constant. A verdict of “no structural change” is therefore not a verdict that diversification held: it locates the *source* of the breakdown in volatility rather than in a regime shift of the dependence structure, and leaves the investor’s experienced diversification exactly as the raw correlations describe it. Read this way the adjustment complements, rather than overturns, the PCA result.

The cleanest benchmark is the `market_spy` specification, because it treats SPY as the common market-shock source. In the common estimation window, SPY’s stress variance is about 3.97 times its calm variance, so the shock used in the adjustment is $\delta_{\text{SPY}} = 2.97$. Figure 4 shows the combined ETF universe pair by pair. Each circle is the raw stress-minus-calm correlation change and each diamond is the same change after the Forbes–Rigobon correction. The pattern is stark: all raw SPY-pair increases that exceed the 0.10 breakdown threshold fall back below it after adjustment. Among the thirteen SPY

pairs, seven are classified as volatility-bias-sensitive increases, three as mostly resilient, three as resilient decreases, and none as robust breakdowns. The largest raw increases therefore describe comovement investors actually experienced in stress, but they do not survive as adjusted correlation increases under the SPY market-shock benchmark.

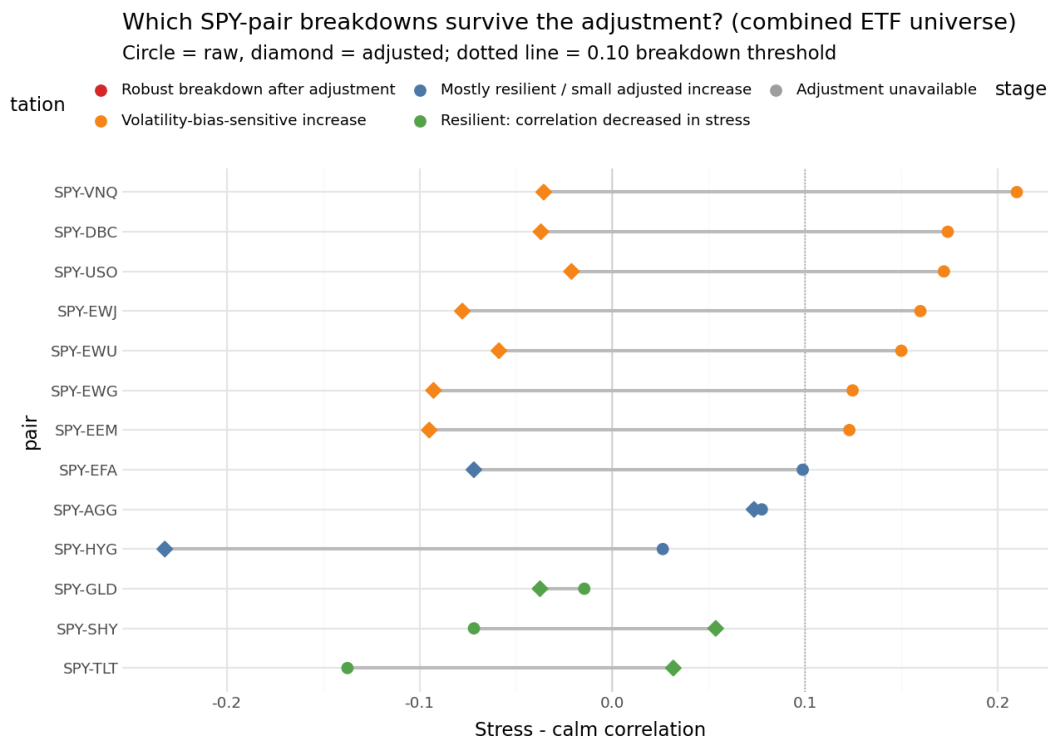


Figure 4: Forbes–Rigobon pair-level survival for the combined ETF universe under the `market_spy` specification. Circles show raw stress-minus-calm correlation changes and diamonds show the adjusted changes after deflating stress correlations by the SPY variance shock. The dotted vertical line marks the 0.10 breakdown threshold. Raw SPY-linked increases that cross the threshold move back below it after correction, so they are volatility-bias-sensitive rather than robust Forbes–Rigobon breakdowns.

Table 5 summarizes the classification across both universes and both adjustment methods. The headline is uniform: *no pair, in any universe or under either method, is a robust breakdown*. The average raw change is positive (or near zero) everywhere but turns negative once adjusted, and the increases that exist in raw space are reclassified as volatility-bias-sensitive—volatility-driven interdependence rather than structural contagion. As the next paragraph shows, the four classes are not spread evenly across the panels: the volatility-bias-sensitive increases are concentrated in international equity, while the cross-asset panel is dominated by genuinely resilient pairs.

The SPY benchmark is methodologically clean but narrow: it asks only whether assets become more correlated with the U.S. equity market. The broader and more revealing evidence comes from the conservative `pair_max` variant restricted to the *non*-SPY pairs, which deflates every pair by the larger of its two assets’ variance shocks—the most de-

Table 5: Forbes–Rigobon pair classification by universe and adjustment method, on the common balanced window. `market_spy` adjusts SPY pairs by the SPY variance shock; `pair_max` adjusts every pair by the larger of its two asset shocks. “Robust” / “Vol.-sens.” / “Resilient” / “Mostly res.” are the four classes of Section 4.6; $\overline{\Delta\rho}$ and $\overline{\bar{\Delta\rho}}$ are the average raw and adjusted stress-minus-calm changes.

Universe	Method	Pairs	Robust	Vol.-sens.	Resilient	Mostly res.	$\overline{\Delta\rho}$	$\overline{\bar{\Delta\rho}}$
Panel A: cross-asset	<code>market_spy</code>	8	0	3	3	2	+0.055	−0.026
Panel A: cross-asset	<code>pair_max</code>	36	0	5	22	9	−0.025	−0.097
Panel B: international equity	<code>market_spy</code>	5	0	4	0	1	+0.131	−0.079
Panel B: international equity	<code>pair_max</code>	15	0	11	0	4	+0.124	−0.068
Combined ETF	<code>market_spy</code>	13	0	7	3	3	+0.084	−0.046
Combined ETF	<code>pair_max</code>	91	0	31	38	22	+0.029	−0.080

manding correction available—and is plotted panel by panel in Figure 5. It is here that the two panels separate most sharply, and the separation is the central diversification result.

In the cross-asset panel the finding is genuine resilience, not merely the absence of contagion. Of the twenty-eight non-SPY pairs, nineteen are *resilient*—their correlation *decreased* in stress—and only two are volatility-bias-sensitive. The diversifiers that matter, Treasuries (SHY, TLT, AGG) and gold paired against the risk assets, co-move *less* in stress even before any adjustment, so they keep diversifying precisely when it is needed; the conservative variance shock only deepens the point. The international-equity panel is the mirror image: *none* of its ten non-SPY pairs is resilient, seven are volatility-bias-sensitive, and the remainder are small increases. Geographic equity diversification therefore produces only the interdependence kind of comovement increase—real for the holder but driven by shared volatility—whereas cross-asset diversification is structurally intact and, for the safe-haven sleeves, actively improves under stress.

Across the combined fourteen-asset universe the same split holds: of the ninety-one pairs none is a robust breakdown, thirty-one are volatility-bias-sensitive (concentrated in the equity block) and thirty-eight are resilient decreases (the cross-asset diversifiers), so the average raw change of +0.029 becomes an adjusted −0.080. The correct reading is therefore not that “the breakdown is not real”—the raw correlations are what an investor experiences—but that the breakdown is *volatility-driven interdependence concentrated in equities*, while the cross-asset hedges genuinely strengthen as diversifiers under stress.

Taken together, the PCA and Forbes–Rigobon evidence sharpen rather than soften the main result. PCA shows that realized diversification concentrates in stress; Forbes–Rigobon shows that this concentration is volatility-driven interdependence rather than structural contagion, and that it is overwhelmingly an equity phenomenon—international-equity pairs supply the volatility-bias-sensitive increases, while the cross-asset hedges remain genuinely resilient. The experienced breakdown is real; its source is heteroskedas-

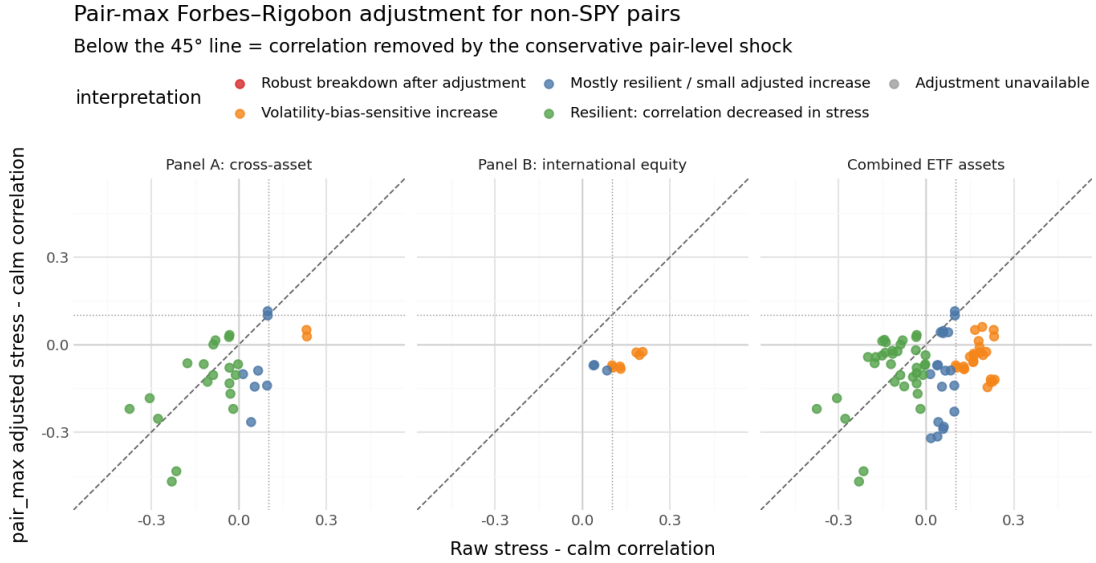


Figure 5: Conservative `pair_max` Forbes–Rigobon adjustment for the non-SPY pairs, by panel. Each point is one pair: the horizontal axis is the raw stress-minus-calm correlation change and the vertical axis the change after deflating by the larger of the two assets’ variance shocks, so points below the 45° line were reduced by the correction. Green points are *resilient* pairs whose correlation *decreased* in stress; orange points are volatility-bias-sensitive increases. The cross-asset panel (left) is dominated by green—Treasures and gold diversify better in stress—while the international-equity panel (centre) contains no green points at all.

ticity, not a regime shift in dependence.

6 Discussion

The three pieces of evidence answer the paper’s question—does diversification fail exactly when it is needed?—in a way that is more precise than a simple “yes”. Read together, they separate the diversification that is genuinely lost in stress from the diversification that only *appears* to be lost because volatility has risen, and they locate where each kind sits.

Diversification does weaken in realized, experienced terms. The raw regime-conditional correlations and the principal-component concentration both move in the fragile direction. In the international-equity panel the whole dependence structure shifts up—the average pairwise correlation rises from 0.74 to 0.87 and the leading component absorbs 89% of variance in stress, leaving barely more than one effective bet across six markets. In the cross-asset panel the average correlation does not rise, yet the portfolio still loses about one-fifth of an independent bet as the risk-on block converges onto a

single direction. Every universe is more concentrated in stress. This is the comovement an investor actually experiences, and on this measure diversification is regime-dependent: a covariance matrix estimated over a full sample understates the joint risk that materializes in a crisis, exactly the instability of the optimization inputs anticipated in the introduction.

The increase is volatility-driven interdependence, and it is concentrated in equities. The Forbes–Rigobon adjustment locates the breakdown rather than dissolving it. Once the stress correlations are deflated for the higher variance of the common factor, *no* pair in any universe survives as a robust breakdown: in the language of Forbes and Rigobon (2002), the evidence points to heightened *interdependence* in a more volatile state rather than *contagion*, a structural break in the dependence relationship. This relabelling does not rescue the investor’s diversification—the experienced comovement is unchanged—but the conservative pair-level correction exposes a sharp asymmetry that the average numbers hide. In the cross-asset panel nineteen of the twenty-eight non-SPY pairs are genuinely resilient, their correlation *falling* in stress, led by Treasuries and gold; the international-equity panel contains no resilient pair and supplies almost all of the volatility-bias-sensitive increases. The honest headline is therefore qualified and specific: diversification weakens in stress, but largely because equity-block variance rises rather than because dependence structurally changes, and the cross-asset hedges keep—indeed strengthen—their diversifying role exactly when it is needed.

A portfolio-management reading. These results are diagnostics for portfolio construction, not allocation rules. The Panel B average shows that adding non-U.S. equity markets does not create independent crisis exposures: the international-equity block tightens almost uniformly, so geographic breadth buys little diversification in precisely the states where it is needed. The Panel A average shows that cross-asset diversification is more robust—but not because every non-equity sleeve turns defensive. Listed real estate, commodities, oil, and high-yield credit remain part of the risk-on complex, while sovereign duration and gold supply the genuinely distinct stress exposures; the VNQ-linked increases are the clearest case, economically consistent with listed real estate behaving as an equity-like risk asset in stress rather than as a separate real-asset hedge. The implication is qualitative rather than a prescription: stress-robust diversification requires genuinely different risk *drivers*, not merely more assets or more countries, and a candidate diversifier is best judged by whether its correlation holds or falls in the stress regime rather than by its unconditional behaviour.

Limitations. Several caveats bound these conclusions. First, the stress sample is small: the regime contains only a handful of distinct episodes, so the regime-conditional estimates rest on limited independent variation. Second, the analysis is associational—the Markov regime is an ex-post label, not an exogenous instrument—so none of the results identify a causal contagion mechanism. Third, the ETF histories constrain the balanced window to begin in 2007, and the universe, while economically representative, is deliberately parsimonious. Fourth, the regime itself is model- and rule-dependent, although the agreement between the Markov and VIX-plus-drawdown labels mitigates this. Fifth, the stress regime pools heterogeneous macroeconomic episodes; the descriptive stress-episode split of Appendix Table 15 shows that hedge signs—most notably the stock–Treasury relation—can differ across stress types. Finally, the Forbes–Rigobon correction is exact only under its single-factor, variance-only assumption and depends on the chosen shock variance; the `market_spy` and `pair_max` variants bracket this choice but do not eliminate it. These limitations qualify the magnitude of the findings, not their direction.

7 Conclusion

This paper asked whether diversification becomes fragile exactly when investors need it, and answered it with three measures conditioned on a stress regime that two independent definitions—a Markov-switching model on returns and a VIX-plus-drawdown rule—agree on. Three findings are robust. First, in realized terms diversification does weaken in stress: pairwise correlations rise and every asset universe concentrates onto fewer independent return directions, most extremely for international equities, whose leading principal component absorbs roughly 89% of variance in stress. Second, the Forbes–Rigobon adjustment shows that this increase is volatility-driven *interdependence* rather than structural *contagion*—no pair, under either adjustment method and in any universe, survives as a robust breakdown. Third, and of most practical relevance, the fragility is concentrated in equities: international-equity pairs supply almost all of the volatility-bias-sensitive increases, while the cross-asset diversifiers—chiefly sovereign duration and gold—are genuinely resilient on average across the pooled stress regime, de-correlating further in most stress episodes, although the descriptive episode split shows the stock–Treasury hedge weakening and turning mildly positive in the inflation-driven 2022–2023 stress (Appendix Table 15); gold’s diversifying role is stable across episode types.

The implication is diagnostic, not prescriptive. Diversification should be evaluated under stress regimes rather than on full-sample averages that blend calm and crisis, because the conditional covariance structure is both more concentrated than the unconditional one and partly inflated by a volatility effect that a naive correlation overstates; and the

exposures that retain diversifying power when correlations rise are different assets, not other equities.

Further research. Three extensions would tighten the evidence. First, *dynamics*: this paper restricts attention to static regime comparisons, so a natural extension is to trace how comovement evolves after a stress onset—using a longer sample or an externally identified volatility shock (for example a high-frequency or narrative instrument) to obtain the statistical power that a handful of ex-post regime transitions cannot provide. Second, *dependence model*: the single-factor, variance-only Forbes–Rigobon correction could be generalized to a multi-factor or dynamic-conditional-correlation specification and complemented by tail-dependence or copula measures, which capture the asymmetric, extreme-state comovement that a Gaussian correlation misses. Third, *economic value*: whether the regime-conditional, variance-adjusted dependence estimates translate into out-of-sample resilience—for instance a regime-aware minimum-variance or risk-parity allocation benchmarked against an unconditional one—is the natural test of whether these diagnostics are useful to an investor rather than only descriptive.

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A Diversification in an Equal-Weighted Portfolio

This appendix expands the portfolio-variance argument in Section 1. Consider the variance of a portfolio containing N risky assets:

$$\sigma_p^2 = \sum_{n=1}^N \sum_{m=1}^N w_n w_m \sigma_{nm}, \quad (18)$$

where $\sigma_{nn} = \sigma_n^2$ is the variance of asset n , while σ_{nm} is the covariance between assets n and m . The covariance matrix therefore contains N diagonal variance terms and $N(N-1)$ ordered off-diagonal covariance terms. Because $\sigma_{nm} = \sigma_{mn}$, the latter correspond to $N(N-1)/2$ unique asset pairs, each appearing twice in Equation (18).

For an equally weighted portfolio, $w_n = 1/N$ for every asset. Separating the diagonal and off-diagonal terms gives

$$\sigma_p^2 = \frac{1}{N^2} \sum_{n=1}^N \sigma_n^2 + \frac{1}{N^2} \sum_{n=1}^N \sum_{\substack{m=1 \\ m \neq n}}^N \sigma_{nm}. \quad (19)$$

Define the average individual variance and average pairwise covariance as

$$\bar{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N \sigma_n^2, \quad \bar{\sigma}_{\text{cov}} = \frac{1}{N(N-1)} \sum_{n=1}^N \sum_{\substack{m=1 \\ m \neq n}}^N \sigma_{nm}. \quad (20)$$

Substitution into Equation (19) yields

$$\boxed{\sigma_p^2 = \frac{1}{N} \bar{\sigma}^2 + \left(1 - \frac{1}{N}\right) \bar{\sigma}_{\text{cov}}}. \quad (21)$$

The scaling explains why adding assets diversifies idiosyncratic risk but does not automatically remove common risk. The N variance terms each receive a weight $1/N^2$, so their combined coefficient is $1/N$. By contrast, the $N(N-1)$ ordered covariance terms have a combined coefficient $1 - 1/N$. Provided the cross-sectional averages remain well behaved as the universe grows,

$$\lim_{N \rightarrow \infty} \sigma_p^2 = \lim_{N \rightarrow \infty} \bar{\sigma}_{\text{cov}}. \quad (22)$$

Thus, the contribution of average individual variance vanishes asymptotically, whereas average covariance determines the remaining portfolio risk.

The role of correlation is especially transparent when assets have a common variance σ^2 and average pairwise correlation $\bar{\rho}$. Because average covariance is then $\bar{\sigma}_{\text{cov}} = \bar{\rho}\sigma^2$,

Equation (21) becomes

$$\sigma_p^2 = \sigma^2 \left[\frac{1}{N} + \left(1 - \frac{1}{N} \right) \bar{\rho} \right] \longrightarrow \bar{\rho} \sigma^2. \quad (23)$$

Increasing N can therefore eliminate asset-specific variance, but it cannot eliminate a positive common dependence component. This is the mathematical motivation for examining whether average correlation and the broader dependence structure become stronger during market stress.

B Asset Universe, Coverage, and Distributional Diagnostics

This appendix collects the asset-universe description of Section 3.1 together with the data-availability table, descriptive return moments, and normal quantile–quantile plots referenced in Section 3.4. Table 6 states the economic role of each cross-asset ETF; the coverage and descriptive tables document raw ETF availability and unconditional daily-return moments; and the figures show the S-shaped departures from the Gaussian benchmark that correspond to the skewness and excess-kurtosis columns of Tables 8–9.

C Regime-Definition Robustness

This appendix collects the robustness material supporting the regime definition of Section 4.1: the model-selection comparison, the agreement between the two independent stress labels, the weekly Markov re-estimation, and the optimizer-initialization check.

D Regime-Conditional Correlations: Supplementary Material

This appendix collects the supporting material for the regime-conditional correlations of Section 5.1: the stress-minus-calm difference heatmaps, the calm-versus-stress scatter plots, the largest pairwise changes, the descriptive-class counts, and the descriptive stress-episode split of selected hedge correlations. All quantities are raw Pearson correlations (Equation (9)) on the common balanced window.

Table 6: Cross-asset universe and economic role

Ticker	Proxy	Role in the stress-diversification design
SPY	U.S. large-cap equities	Baseline risky market portfolio and benchmark equity beta; tracks the S&P 500.
SHY	Short U.S. Treasuries	Cash-like, short-duration government bond sleeve; captures the defensive short end of the Treasury curve.
TLT	Long U.S. Treasuries	Long-duration government bond sleeve; captures duration risk and the classic flight-to-quality hedge channel.
AGG	U.S. aggregate bonds	Broad investment-grade fixed-income market; benchmark for core bond exposure.
HYG	U.S. high-yield credit	Risky corporate credit; sits between equities and safer bonds and is especially useful in stress episodes.
GLD	Gold	Precious-metals safe-haven sleeve and classic crisis diversifier candidate.
DBC	Broad commodities	Diversified commodity basket; captures inflation- and real-asset exposure beyond precious metals.
USO	Crude oil	Concentrated energy exposure; useful because oil often behaves differently from both broad commodities and financial assets in stress episodes.
VNQ	Listed U.S. real estate	Equity-like real-asset sleeve; allows the analysis to distinguish listed property risk from broad equity beta.

Table 7: Asset-level data availability in the raw daily ETF dataset

Panel	Ticker	First observation	Number of daily observations
Cross-asset	SPY	2000-01-03	6,538
Cross-asset	SHY	2002-07-30	5,894
Cross-asset	TLT	2002-07-30	5,894
Cross-asset	AGG	2003-09-29	5,600
Cross-asset	VNQ	2004-09-29	5,348
Cross-asset	GLD	2004-11-18	5,312
Cross-asset	DBC	2006-02-06	5,007
Cross-asset	USO	2006-04-10	4,963
Cross-asset	HYG	2007-04-11	4,712
International equity	EWG	2000-01-03	6,538
International equity	EWJ	2000-01-03	6,538
International equity	EWU	2000-01-03	6,538
International equity	SPY	2000-01-03	6,538
International equity	EFA	2001-08-27	6,122
International equity	EEM	2003-04-14	5,716

Table 8: Descriptive statistics of daily log returns for the cross-asset panel, each ETF over its full availability window (N counts daily returns)

Ticker	N	Mean (ann., %)	Std. dev. (ann., %)	Skewness	Excess kurtosis
AGG	5,599	3.10	5.21	-2.05	54.57
DBC	5,006	0.83	19.22	-0.49	3.35
GLD	5,311	10.42	17.67	-0.34	5.87
HYG	4,711	4.87	10.99	0.31	38.93
SHY	5,893	1.98	1.52	0.28	7.10
SPY	6,537	7.77	19.40	-0.21	11.54
TLT	5,893	3.74	14.38	-0.02	3.40
USO	4,962	-10.43	36.76	-1.10	13.93
VNQ	5,347	7.08	28.70	-0.54	18.73

Table 9: Descriptive statistics of daily log returns for the international equity panel, each ETF over its full availability window (N counts daily returns)

Ticker	N	Mean (ann., %)	Std. dev. (ann., %)	Skewness	Excess kurtosis
EEM	5,715	8.93	27.01	0.02	16.15
EFA	6,121	6.05	20.86	-0.32	12.66
EWG	6,537	4.35	25.48	-0.22	9.01
EWJ	6,537	2.52	21.38	-0.06	7.32
EWU	6,537	3.79	22.47	-0.51	12.70
SPY	6,537	7.77	19.40	-0.21	11.54

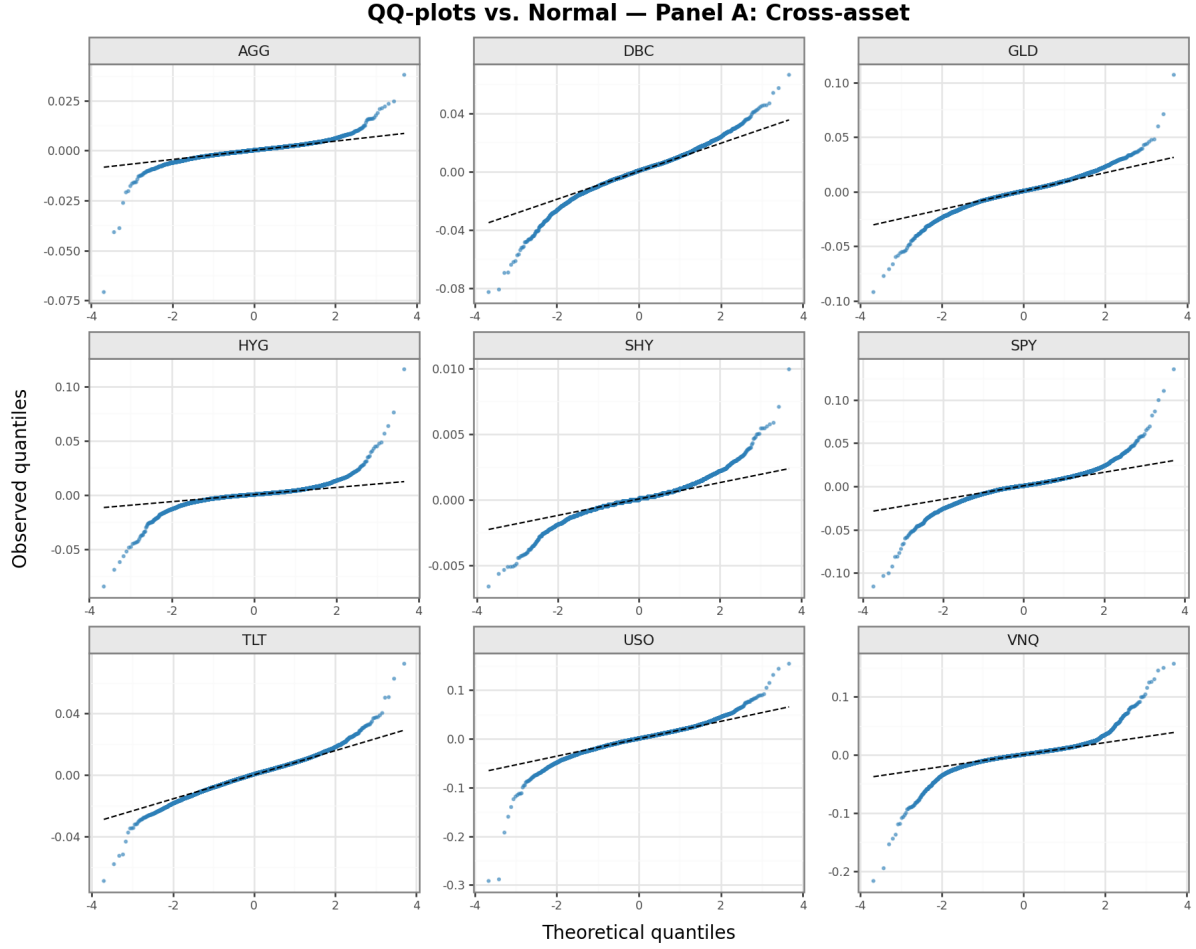


Figure 6: Normal Q–Q plots for daily log returns in Panel A. The dashed line represents the Gaussian benchmark. Systematic departures in the tails indicate that observed ETF return quantiles are more extreme than the corresponding normal quantiles.

Table 10: Model selection for the monthly Markov-switching model. Lower BIC is preferred; the chosen specification is in bold.

States	Switching parameters	Log-likelihood	AIC	BIC
2	mean and variance (official)	−870.6	1753.2	1775.6
2	variance only	−874.5	1758.9	1777.6
2	mean only	−886.0	1782.0	1800.7
3	mean and variance	−855.9	1735.8	1780.7
3	variance only	−872.4	1764.8	1802.2

The three-state specifications did not converge to a stable optimum and are reported for completeness only.

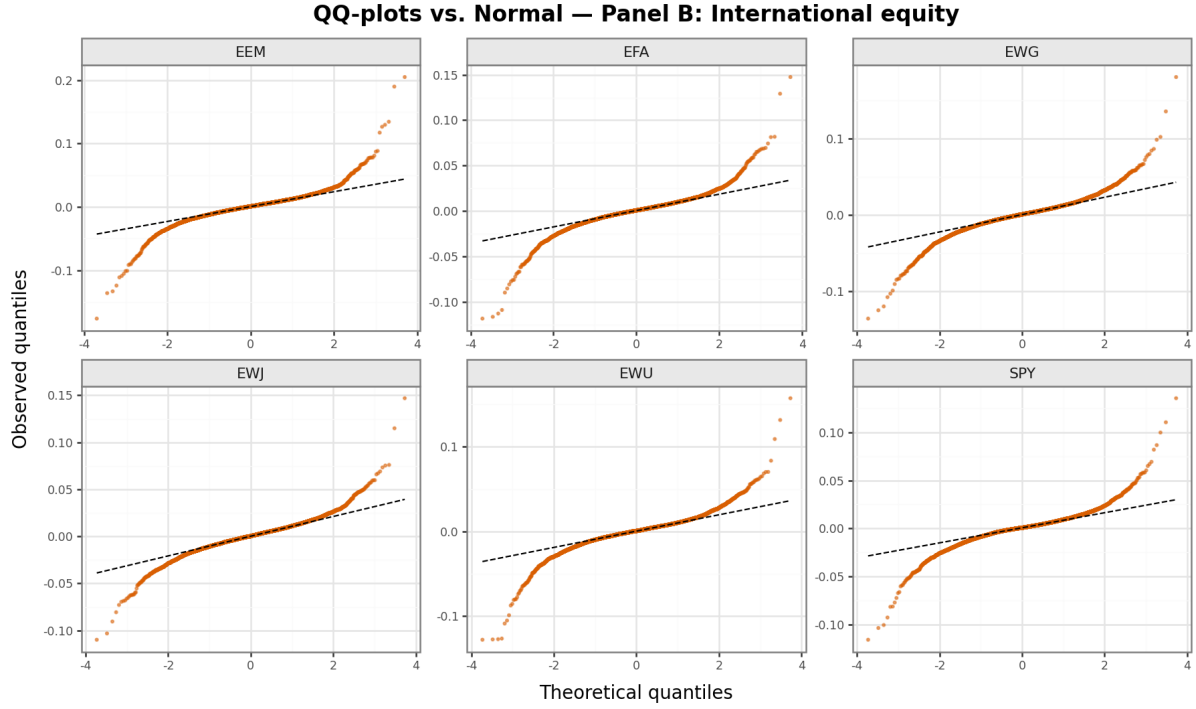


Figure 7: Normal Q–Q plots for daily log returns in Panel B. The repeated S-shaped deviations from the Gaussian benchmark indicate non-normal tail behavior across the international equity ETFs.

Table 11: Agreement between the Markov stress label and the rule-based VIX-plus-drawdown label, measured by Cohen’s κ . The last row compares the weekly and monthly Markov labels with each other.

Comparison	Frequency / threshold	Agreement	Cohen’s κ
Markov vs. rule-based	Monthly, 0.50	65.1%	0.26
Markov vs. rule-based	Monthly, 0.75	71.3%	0.31
Markov vs. rule-based	Weekly, 0.50	83.9%	0.52
Markov vs. rule-based	Weekly, 0.75	86.4%	0.54
Weekly vs. monthly Markov	0.50	75.4%	0.49

Monthly Markov-switching SPY regime · two-state, switching mean and variance · stress = high-variance state (47% of days)

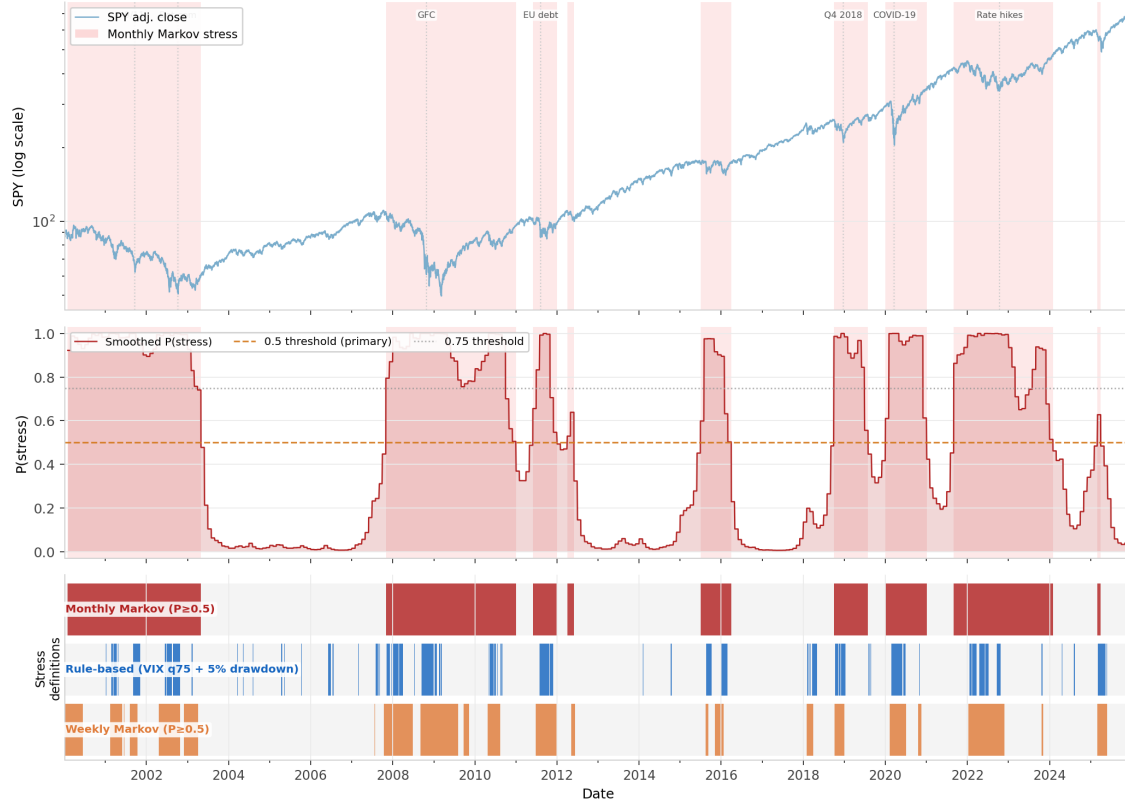


Figure 8: Monthly Markov-switching SPY regime. Top: SPY (log scale) with the identified stress months shaded. Middle: smoothed stress probability with the 0.5 and 0.75 classification thresholds. Bottom: regime “swim-lanes” comparing the monthly Markov stress label against the rule-based VIX-plus-drawdown label and the weekly Markov label.

Table 12: Optimizer-initialization robustness of the monthly Markov-switching model. The official model is re-estimated with additional EM warm-up iterations and twenty random starting-value searches (`em_iter=20`, `search_reps=20`, `search_iter=10`, `maxiter=1000`) and compared with the default fit. The two fits are numerically indistinguishable.

Quantity	Default (official)	Robust search
Log-likelihood	−870.6	−870.6
AIC	1753.1	1753.1
BIC	1775.6	1775.6
Stress-state variance	31.93	31.93
Calm-state variance	5.67	5.67
Stress persistence p_{ss}	0.939	0.939
Stress share (≥ 0.5)	0.473	0.473

Smoothed stress-probability paths correlate at $\rho \approx 1.00$; the 0.5 stress labels agree in all 311 months.

Weekly Markov-switching SPY regime · two-state, switching mean and variance · stress = high-variance state (26% of days)

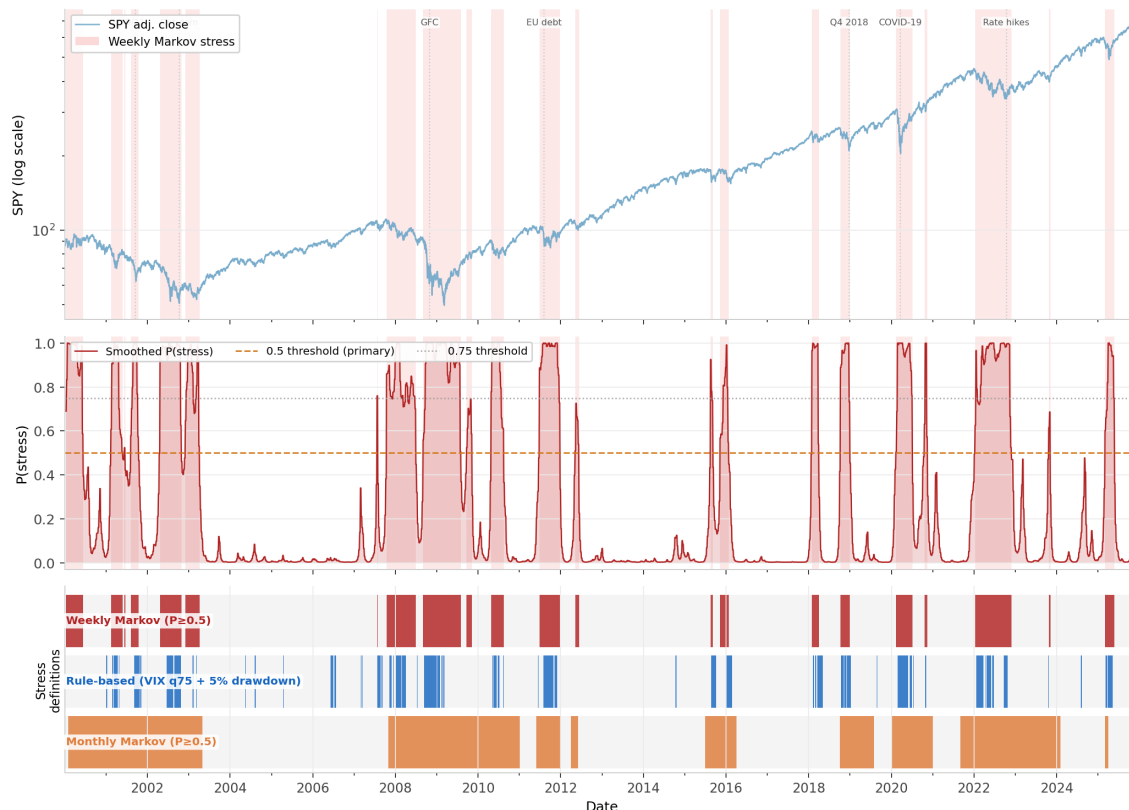


Figure 9: Weekly Markov-switching SPY regime (robustness check), in the same layout as Figure 8. The weekly model flags fewer periods as stress and tracks the rule-based label more closely, while still identifying the same crisis episodes.

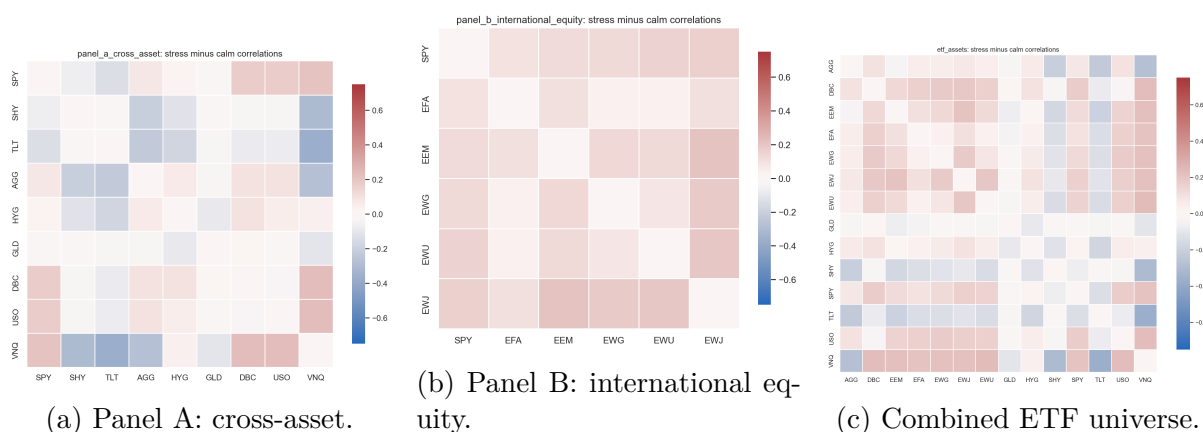


Figure 10: Stress-minus-calm correlation difference matrices $\Delta\rho_{ij}$ by universe. Warm cells mark pairs that tighten in stress and cool cells mark pairs that decouple. Panel B is uniformly warm; Panel A shows the reshuffle—a warm risk-on (equity–real-estate–commodity) block and a cool safe-haven (Treasury–gold) block.

Table 13: Largest pairwise correlation changes by universe (raw Pearson, common balanced window). Each block lists the most extreme stress-minus-calm changes; Panel B has no pair whose correlation decreases, so its smallest increases are shown instead. Complete rankings for all 36, 15, and 91 pairs are in the project outputs (*_pair_ranking.csv).

Pair	ρ^{calm}	ρ^{stress}	$\Delta\rho$
<i>Panel A: cross-asset — largest increases</i>			
DBC–VNQ	0.108	0.341	+0.233
USO–VNQ	0.067	0.298	+0.231
SPY–VNQ	0.585	0.795	+0.210
SPY–DBC	0.294	0.468	+0.174
SPY–USO	0.247	0.419	+0.172
<i>Panel A: cross-asset — largest decreases</i>			
TLT–VNQ	0.123	−0.252	−0.374
SHY–VNQ	0.108	−0.198	−0.306
AGG–VNQ	0.240	−0.037	−0.277
TLT–AGG	0.877	0.647	−0.230
SHY–AGG	0.752	0.538	−0.214
<i>Panel B: international equity — largest increases</i>			
EEM–EWJ	0.596	0.802	+0.206
EWU–EWJ	0.605	0.800	+0.195
EWG–EWJ	0.613	0.797	+0.184
SPY–EWJ	0.657	0.816	+0.160
SPY–EWU	0.718	0.868	+0.150
<i>Panel B: international equity — smallest increases (no decreases)</i>			
EFA–EWG	0.912	0.949	+0.038
EFA–EWU	0.908	0.948	+0.041
EWG–EWU	0.812	0.896	+0.084
SPY–EFA	0.810	0.909	+0.099
EFA–EEM	0.799	0.900	+0.101
<i>Combined ETF universe — largest increases</i>			
VNQ–EWU	0.479	0.714	+0.234
DBC–VNQ	0.108	0.341	+0.233
USO–VNQ	0.067	0.298	+0.231
VNQ–EEM	0.487	0.715	+0.228
VNQ–EWG	0.464	0.686	+0.222
<i>Combined ETF universe — largest decreases</i>			
TLT–VNQ	0.123	−0.252	−0.374
SHY–VNQ	0.108	−0.198	−0.306
AGG–VNQ	0.240	−0.037	−0.277
TLT–AGG	0.877	0.647	−0.230
SHY–AGG	0.752	0.538	−0.214

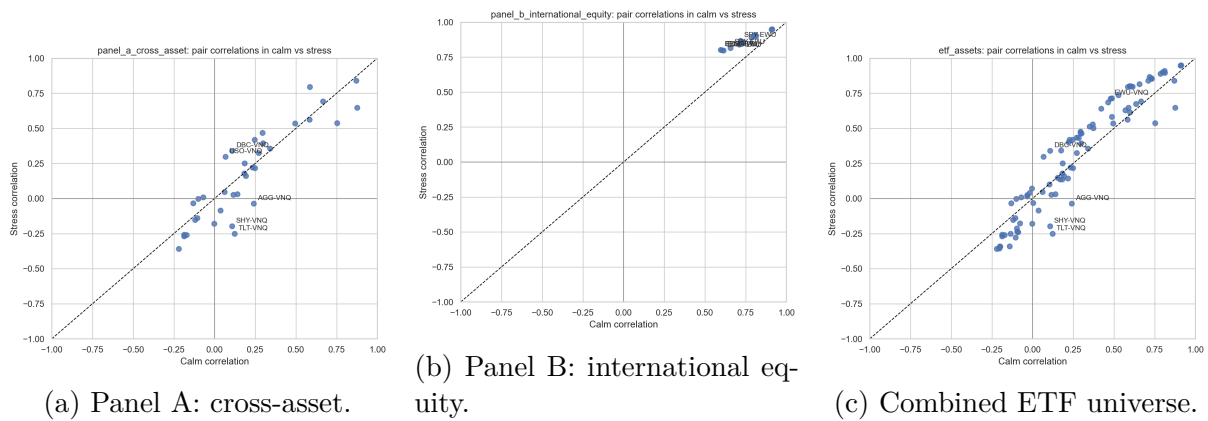


Figure 11: Calm (horizontal) versus stress (vertical) pairwise correlation by universe. Points above the 45° line tighten in stress; points below it decouple. Panel B lies entirely above the line, while Panel A splits across it.

Table 14: Descriptive-class counts of the pairwise correlation change by universe, on the common balanced window. *Resilient* pairs decouple in stress ($\Delta\rho < 0$); *small increase* pairs rise but stay below the 0.10 breakdown threshold; *breakdown* pairs cross it ($\Delta\rho \geq 0.10$). Shares are of the universe’s pair count.

Universe	Resilient ($\Delta\rho < 0$)	Small increase (0–0.10)	Breakdown (≥ 0.10)	Total
Panel A: cross-asset	22 (61.1%)	9 (25.0%)	5 (13.9%)	36
Panel B: international equity	0 (0.0%)	4 (26.7%)	11 (73.3%)	15
Combined ETF universe	38 (41.8%)	22 (24.2%)	31 (34.1%)	91

Table 15: Stress-episode heterogeneity of selected hedge correlations. Correlations are computed only on Markov-stress days within each broad era, using the same common balanced return window as the main analysis. The pooled-stress row corresponds to the static stress-regime estimate used in the body. The split is descriptive and is not used for formal inference.

Episode	Stress days	Share	SPY–TLT	SPY–AGG	SPY–GLD	SPY–VNQ	TLT–AGG	TLT–VNQ
Pooled stress	2,268	100.0%	−0.358	+0.008	+0.046	+0.795	+0.647	−0.252
2007–2019	1,387	61.2%	−0.482	−0.110	+0.010	+0.795	+0.508	−0.349
2020–2021	338	14.9%	−0.452	+0.130	+0.145	+0.877	+0.640	−0.383
2022–2023	501	22.1%	+0.099	+0.262	+0.130	+0.774	+0.914	+0.265

“Share” is the era’s share of the 2,268 Markov-stress days in the balanced window. The 2024–2025 period contains only 42 stress days (1.9% of stress days), below the 100-day reporting threshold, and is omitted. The full per-era, per-pair tables and the heterogeneity ranking are exported by notebook 03 to `outputs/tables/`.

E Bootstrap Inference: Supplementary Material

This appendix collects the supporting material for the bootstrap inference of Section 5.2: the bootstrap sampling distributions of the universe averages, the full set of pairwise confidence intervals for the top and tail pairs, the inferential detail for the most extreme pairs, the block-length sensitivity check, and the comparison with the classical Fisher- z test. Unless noted, results use $B = 2,000$ circular moving-block replicates with block length $L = 21$ days (Section 4.3).

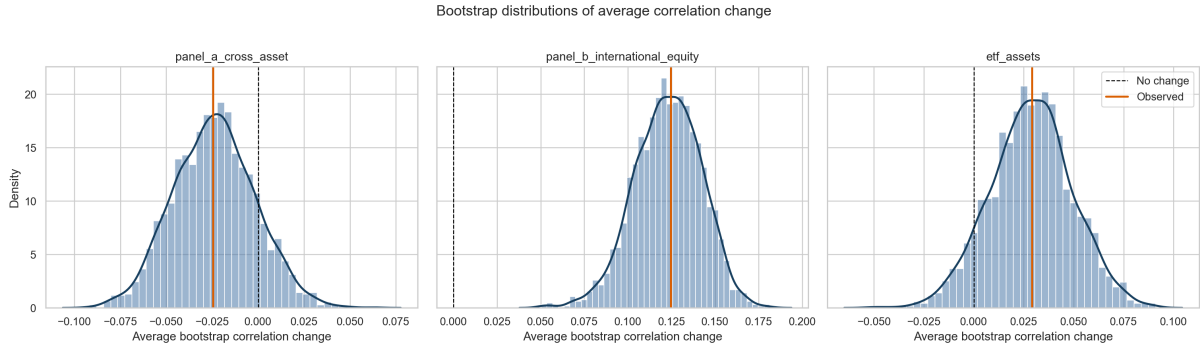


Figure 12: Bootstrap sampling distributions of the universe-average correlation change $\overline{\Delta\rho}$, one panel per universe. The dashed line marks the no-change value of zero and the solid line the observed point estimate. Only the international-equity distribution is cleanly separated from zero; the cross-asset and combined distributions straddle it, the visual counterpart of the insignificant averages in Table 3.

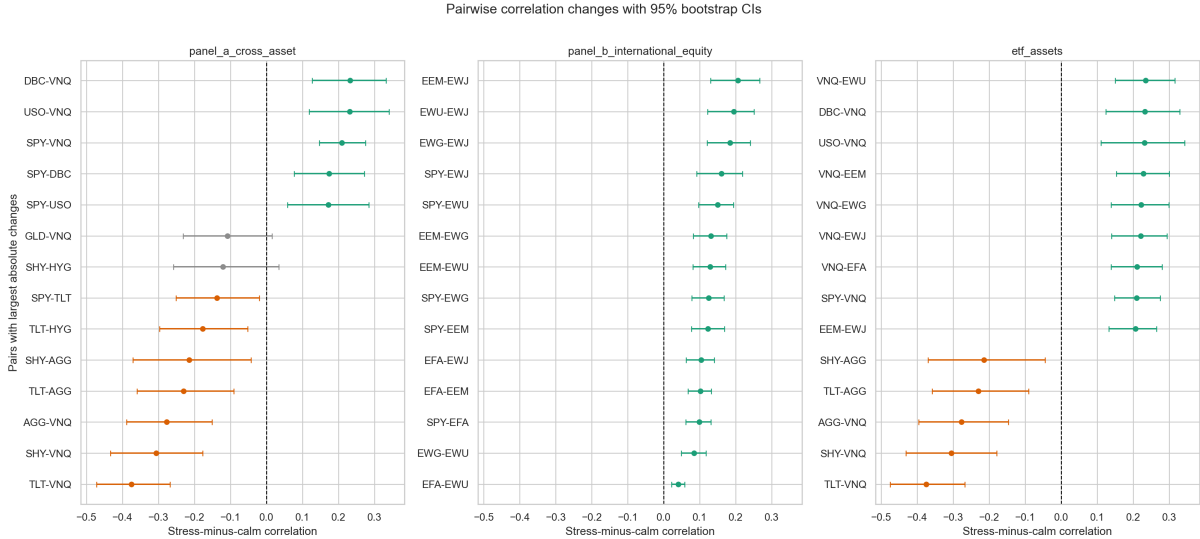


Figure 13: Pairwise stress-minus-calm correlation changes with 95% moving-block bootstrap confidence intervals, for the largest-magnitude pairs in each universe (top and tail). Pairs whose interval excludes zero are significant increases (above zero) or decreases (below zero); intervals crossing zero are inconclusive. This is the pair-level detail behind the significance counts of Table 3.

Table 16: Bootstrap inference for the most extreme pairs of the combined ETF universe (largest increases and largest decreases). $\Delta\rho$ is the raw change; the 95% interval and two-sided p -value (Equation (11)) are from the moving-block bootstrap; q_{BH} is the Benjamini–Hochberg false-discovery-rate-adjusted value. All listed pairs are FDR-significant ($q_{\text{BH}} < 0.05$).

Pair	$\Delta\rho$	95% CI	p_{boot}	q_{BH}
<i>Largest increases</i>				
VNQ–EWU	+0.234	[+0.150, +0.315]	< 0.001	0.002
DBC–VNQ	+0.233	[+0.124, +0.329]	< 0.001	0.002
USO–VNQ	+0.231	[+0.110, +0.343]	< 0.001	0.002
VNQ–EEM	+0.228	[+0.153, +0.300]	< 0.001	0.002
VNQ–EWG	+0.222	[+0.139, +0.299]	< 0.001	0.002
VNQ–EWJ	+0.221	[+0.139, +0.294]	< 0.001	0.002
VNQ–EFA	+0.210	[+0.138, +0.281]	< 0.001	0.002
SPY–VNQ	+0.210	[+0.148, +0.275]	< 0.001	0.002
<i>Largest decreases</i>				
TLT–VNQ	−0.374	[−0.475, −0.267]	< 0.001	0.002
SHY–VNQ	−0.306	[−0.431, −0.179]	< 0.001	0.002
AGG–VNQ	−0.277	[−0.395, −0.147]	< 0.001	0.002
TLT–AGG	−0.230	[−0.358, −0.090]	< 0.001	0.002
SHY–AGG	−0.214	[−0.370, −0.045]	0.004	0.010
TLT–EEM	−0.199	[−0.300, −0.101]	< 0.001	0.002
TLT–HYG	−0.176	[−0.299, −0.050]	0.008	0.018
SHY–EEM	−0.172	[−0.301, −0.042]	0.007	0.016

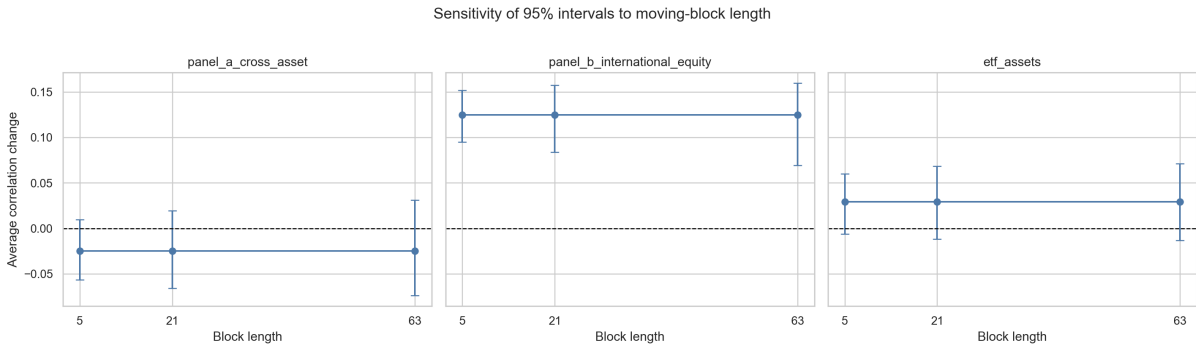


Figure 14: Block-length sensitivity of the universe-average correlation change. Point estimates are invariant to the block length; the 95% intervals widen as the block lengthens from 5 to 63 days, the expected effect of acknowledging more serial dependence. The international-equity average stays significant at every block length.

Table 17: Block-length sensitivity of the universe-average correlation change. Point estimates do not depend on the block length L ; the 95% bootstrap interval widens with L , and the FDR-significant pair count falls. The $L = 21$ row is the primary specification (Table 3); the $L \in \{5, 63\}$ rows use 500 replicates.

Universe	L	$\overline{\Delta\rho}$	95% CI	FDR-sig. pairs
Panel A: cross-asset	5	-0.025	$[-0.056, +0.010]$	12
	21	-0.025	$[-0.066, +0.019]$	11
	63	-0.025	$[-0.074, +0.031]$	9
Panel B: international equity	5	+0.124	$[+0.095, +0.152]$	15
	21	+0.124	$[+0.084, +0.157]$	15
	63	+0.124	$[+0.069, +0.160]$	15
Combined ETF universe	5	+0.029	$[-0.006, +0.060]$	54
	21	+0.029	$[-0.012, +0.068]$	50
	63	+0.029	$[-0.013, +0.071]$	40

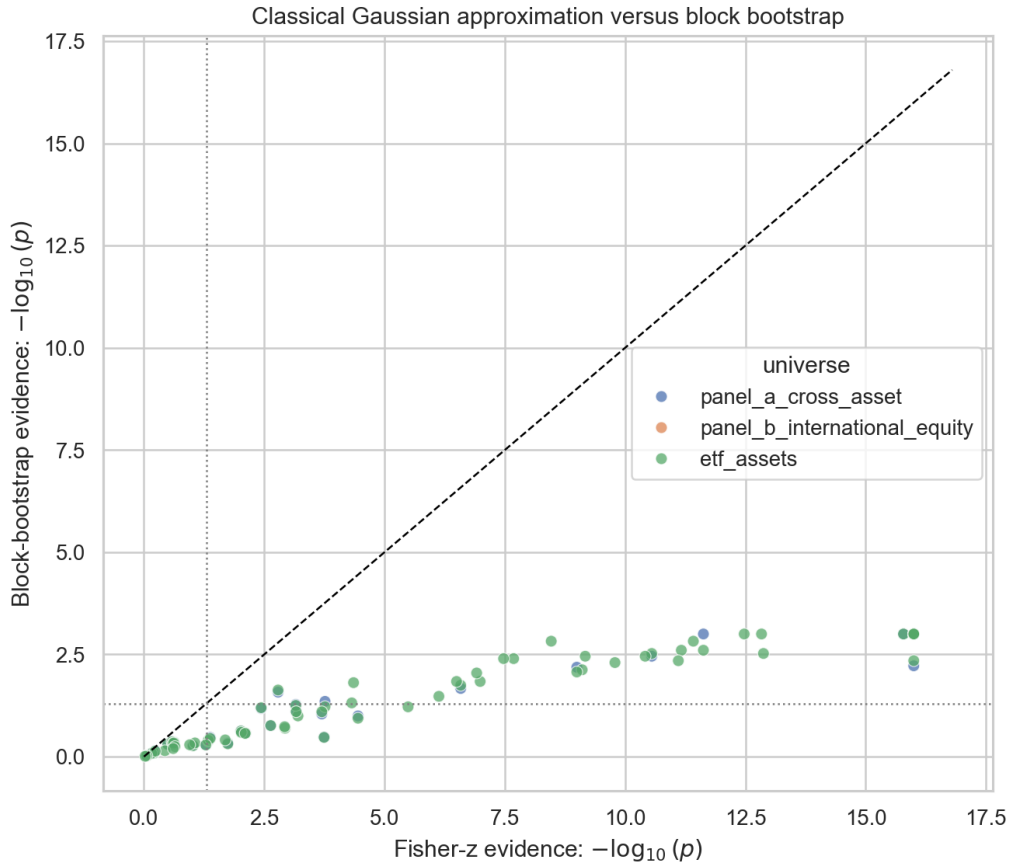


Figure 15: Block-bootstrap versus classical Fisher-z evidence against the null of no correlation change, one point per pair (plotted as $-\log_{10} p$). Points below the 45° line are pairs the bootstrap finds less significant than Fisher-z. The systematic shift quantifies the over-rejection induced by the i.i.d. Gaussian assumption on serially dependent daily returns.

Table 18: Agreement between the block-bootstrap (FDR-controlled, $q < 0.05$) and the unadjusted classical Fisher- z test in declaring a pair’s correlation change significant, by universe. “Both” counts pairs significant under both; “Fisher only” counts pairs significant under Fisher- z but not the bootstrap. No pair is significant under the bootstrap alone.

Universe	Pairs	Bootstrap (FDR)	Fisher- z	Both	Bootstrap only	Fisher only	Neither
Panel A: cross-asset	36	11	25	11	0	14	11
Panel B: international equity	15	15	15	15	0	0	0
Combined ETF universe	91	50	73	50	0	23	18

F Principal-Component Concentration: Supplementary Figures

This appendix collects the supporting figures for the principal-component concentration of Section 5.3. Figure 16 places each universe in the $(\Delta \text{pc1_share}, \Delta N_{\text{eff}})$ plane; all three lie in the upper-left “more concentrated in stress” region (a higher leading-component share together with fewer effective bets). Figures 17–18 report the full eigenvalue spectra and cumulative explained-variance curves by regime for each panel.

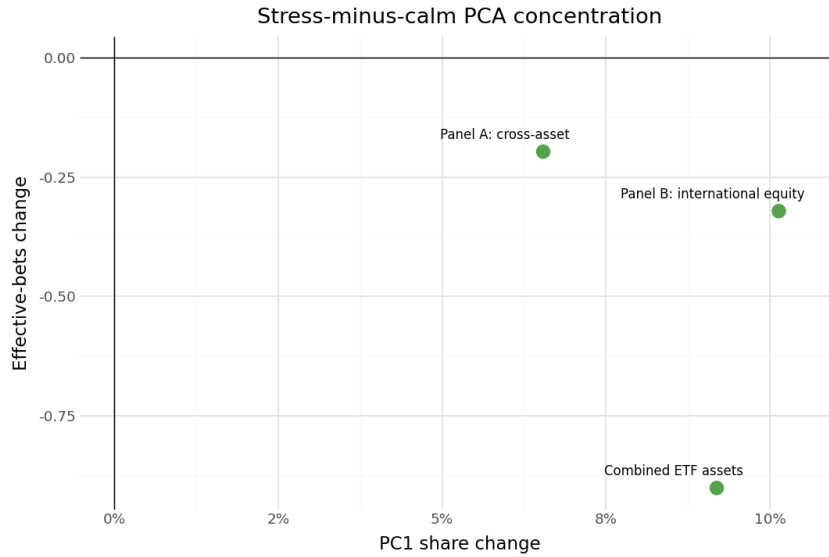


Figure 16: Stress-minus-calm change in the leading-component share (x -axis) against the change in the effective number of bets (y -axis), one marker per universe. All three universes sit in the upper-left region— pc1_share rises and N_{eff} falls—confirming the concentration verdict of Table 4.

Table 19: Panel B (international equity) principal-component concentration: common balanced window (used in the body) versus the full sample. Panel B is the only universe the common-window restriction shortens; it contains no asset with a 2007 inception, so it can in principle start in 2003. The concentration verdict—a higher pc1_share and a lower N_{eff} in stress—holds on both windows, so the body’s harmonization to the common window is innocuous for the conclusion.

Window	Obs. (calm/stress)	pc1_share		N_{eff}	
		Calm	Stress	Calm	Stress
Common (2007–2025, body)	2,443 / 2,268	0.787	0.889	1.58	1.26
Full sample (2003–2025)	3,436 / 2,279	0.779	0.888	1.62	1.26

Both windows satisfy the verdict ($\text{pc1_share}^{\text{stress}} > \text{pc1_share}^{\text{calm}}$ and $N_{\text{eff}}^{\text{stress}} < N_{\text{eff}}^{\text{calm}}$). Restricting Panel B to the common window drops roughly a thousand mostly-calm 2003–2007 trading days, which is why its calm pc1_share rises slightly while the stress figures are essentially unchanged.

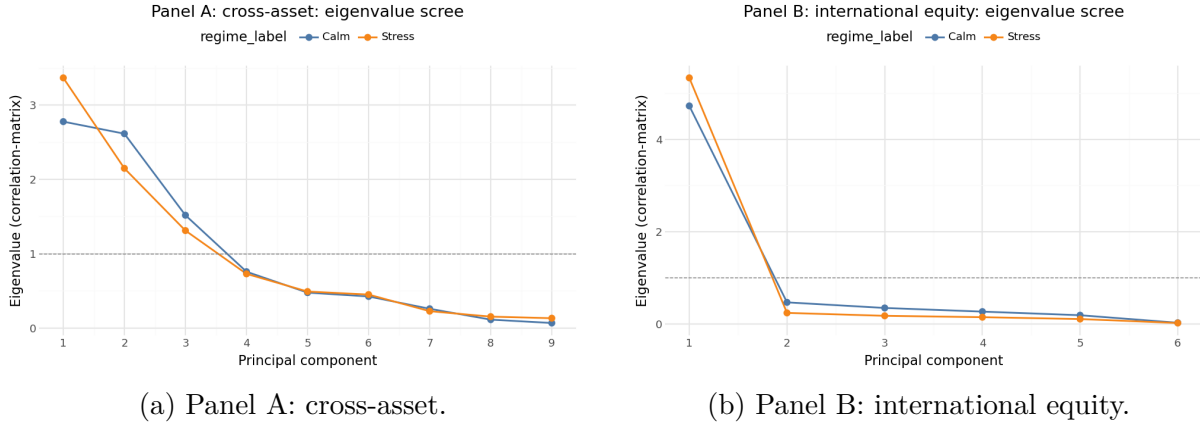


Figure 17: Scree plots of the regime-conditional correlation eigenvalues, calm versus stress. In stress the leading eigenvalue rises and the spectrum steepens—variance reallocates into the first component—most starkly for the international-equity panel.

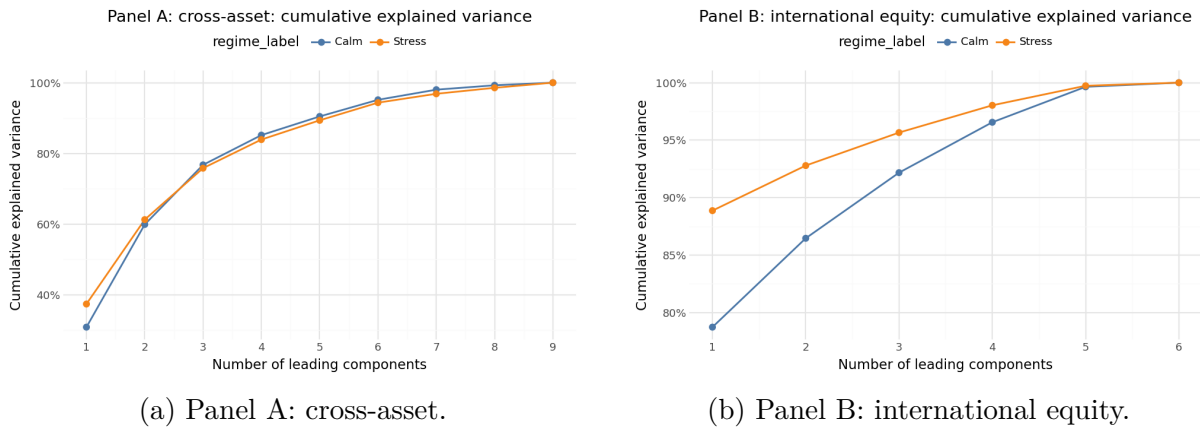


Figure 18: Cumulative explained-variance curves by regime. The stress curve lies above the calm curve at the first component for both panels, the cumulative-share statement of the rising leading-component concentration.

G Forbes–Rigobon Robustness Variants

The body reports the main `market_spy` adjustment and the class counts for the conservative `pair_max` variant (Table 5). Table 20 adds the full `pair_max` distribution—average *and* median raw and adjusted changes—for each universe. The pattern is the same everywhere: a positive (or near-zero) raw stress-minus-calm change becomes a negative adjusted change once each pair is deflated by the larger of its two asset variance shocks, and not a single pair survives as a robust breakdown.

Table 20: Forbes–Rigobon `pair_max` variant: distribution of raw and adjusted stress-minus-calm correlation changes by universe, on the common balanced window. $\Delta\rho$ is the raw change and $\tilde{\Delta}\rho$ the change after deflating the stress correlation by $\max(\delta_i, \delta_j)$. “Robust” counts pairs with both $\Delta\rho \geq 0.10$ and $\tilde{\Delta}\rho \geq 0.10$.

Universe	Pairs	$\Delta\rho$		$\tilde{\Delta}\rho$		Robust
		Mean	Median	Mean	Median	
Panel A: cross-asset	36	−0.025	−0.024	−0.097	−0.084	0
Panel B: international equity	15	+0.124	+0.125	−0.068	−0.072	0
Combined ETF	91	+0.029	+0.050	−0.080	−0.068	0